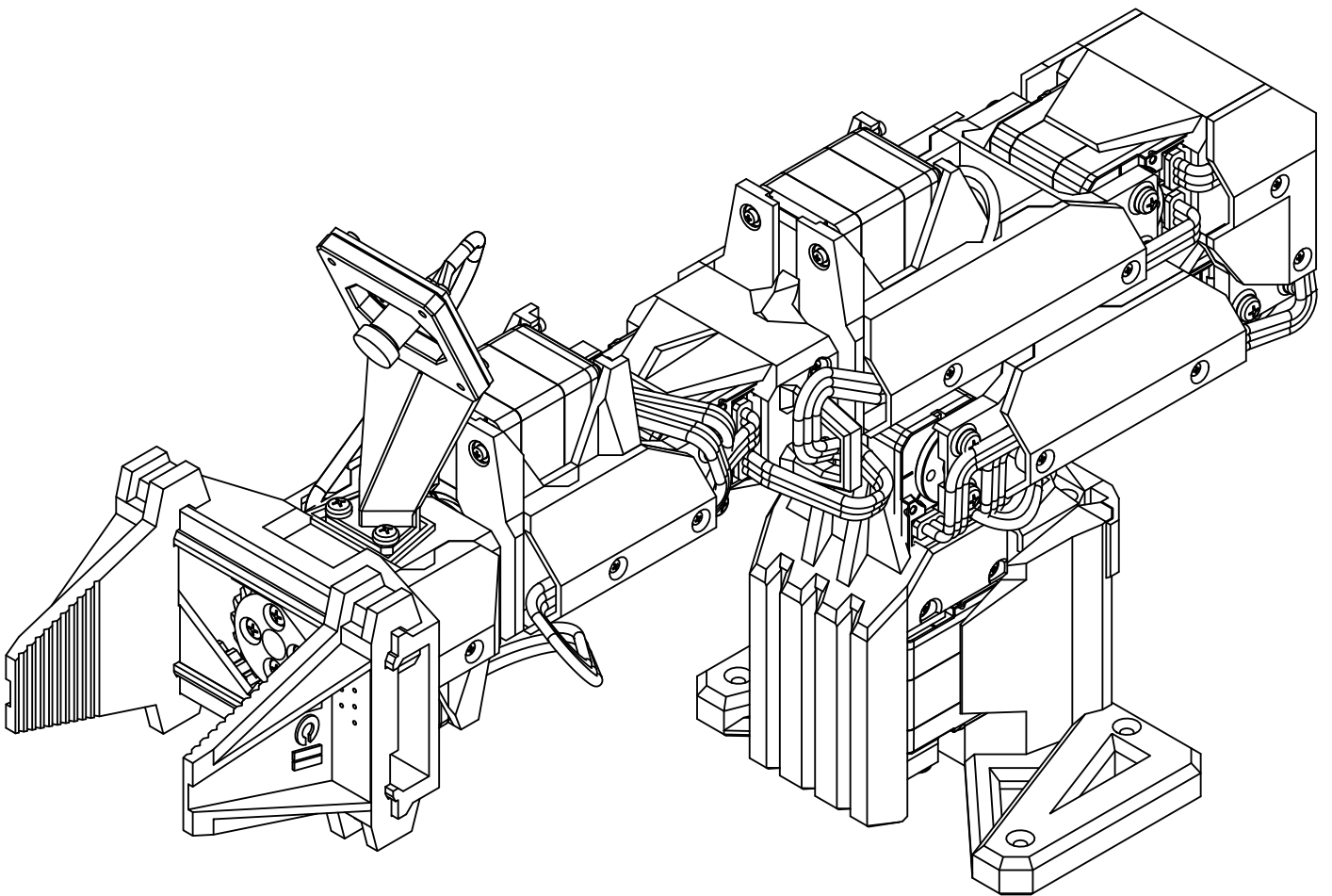


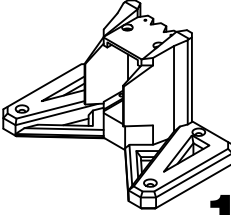
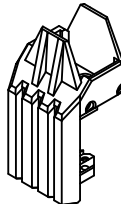
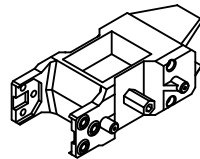
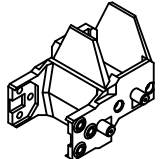
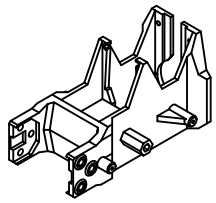
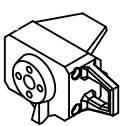
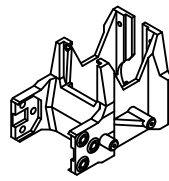
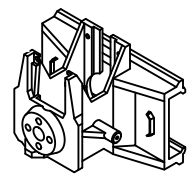
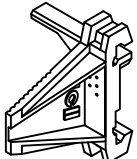
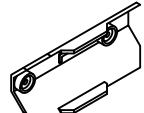
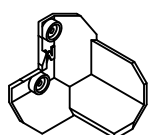
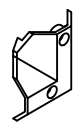
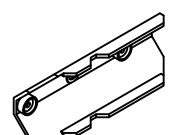
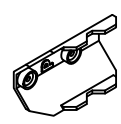
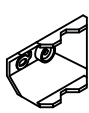
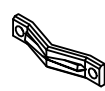
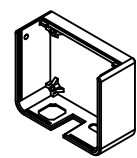
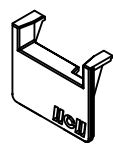
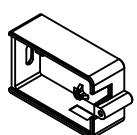
ELROBOT

[HTTPS://NORMACORE.DEV/](https://normacore.dev/)



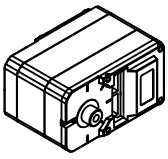
ELROBOT

3D PRINTING PARTS

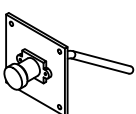
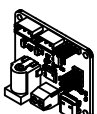
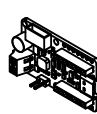
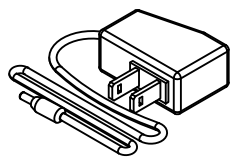
BASE  A 1x	JOINT 1  B 1x	JOINT 2  C 1x	JOINT 3  D 1x
JOINT 4  E 1x	JOINT 5  F 1x	JOINT 6  G 1x	GRIPPER BASE  H 1x
GRIPPER GEAR  I 1x	GRIPPER JAW  J 2x	BASE SHIELD  K 1x	JOINT 2 SHIELD  L 2x
JOINT 3 SHIELD DRIVE SIDE  M 1x	JOINT 3 SHIELD CAMERA SIDE  N 1x	JOINT 4 SHIELD  O 2x	JOINT 6 SHIELD  P 2x
GRIPPER BASE SHIELD  Q 2x	WIRING BRACKET  R 1x	CAMERA MOUNT  S 1x	ENCLOSURE  T1 1x
ENCLOSURE  T2 1x	ENCLOSURE  U1 1x	ENCLOSURE  U2 1x	

ELROBOT

SERVO PARTS

SERVO ST3215  V1 8x	IDLER HORN  V2 6x	SERVO HORN  V3 8x	SERVO WIRES  V4 8x
SCREW M3x8  V5 54x	SELF-TAPPING SCREW M2x6  V6 61x		

ADDITIONAL PARTS

UVC CAMERA INNOMAKER U20CAM  W1 1x	SERVO DRIVER BOARD  W2 1x	SERVO DRIVER BOARD  W3 1x	POWER SUPPLY  W4 1x
--	--	---	--

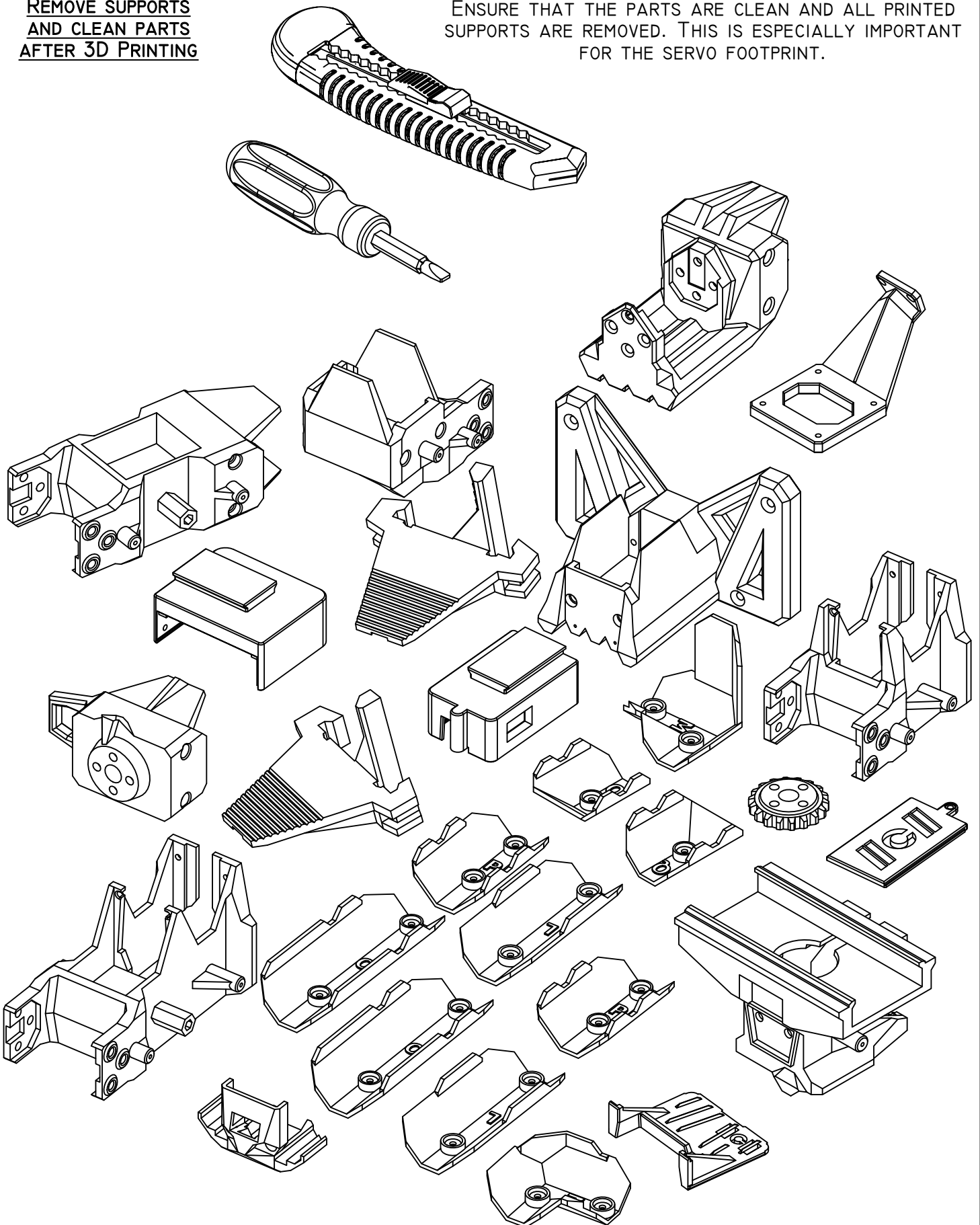
To assemble ElRobot you will need:
screwdrivers PH1 and PH2,
a marker or pencil,
a pair of clamps (you could print it),
a stationery knife
and a good mood.
Good luck and have fun!

ELROBOT

STEP 1

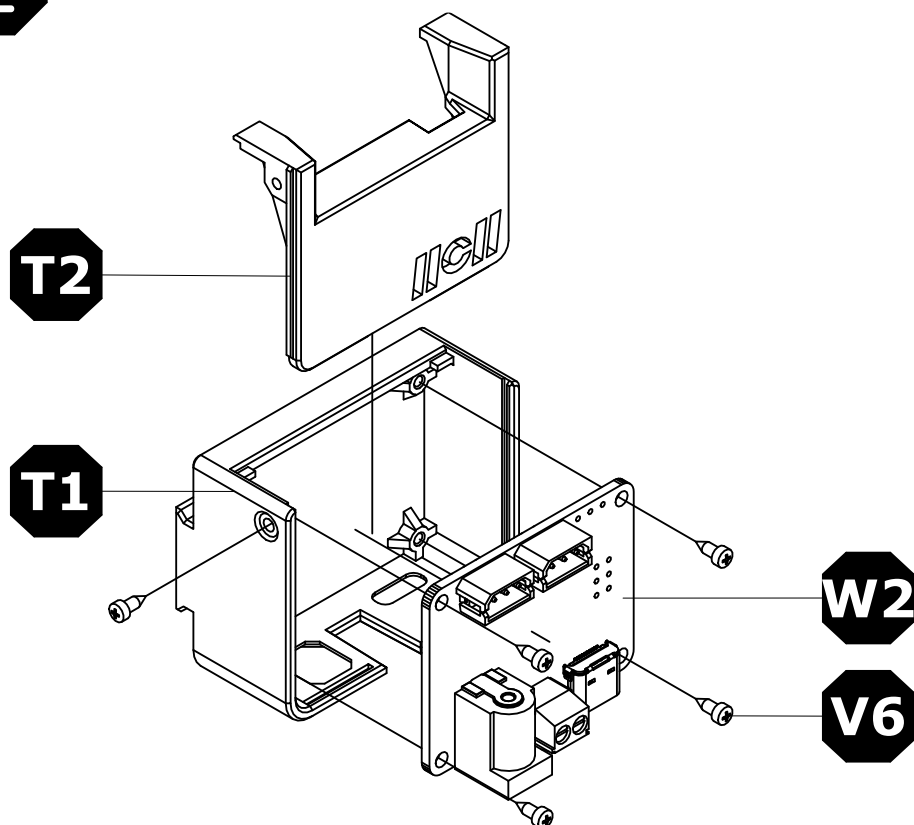
REMOVE SUPPORTS
AND CLEAN PARTS
AFTER 3D PRINTING

ENSURE THAT THE PARTS ARE CLEAN AND ALL PRINTED
SUPPORTS ARE REMOVED. THIS IS ESPECIALLY IMPORTANT
FOR THE SERVO FOOTPRINT.



ELROBOT

OPTIONAL



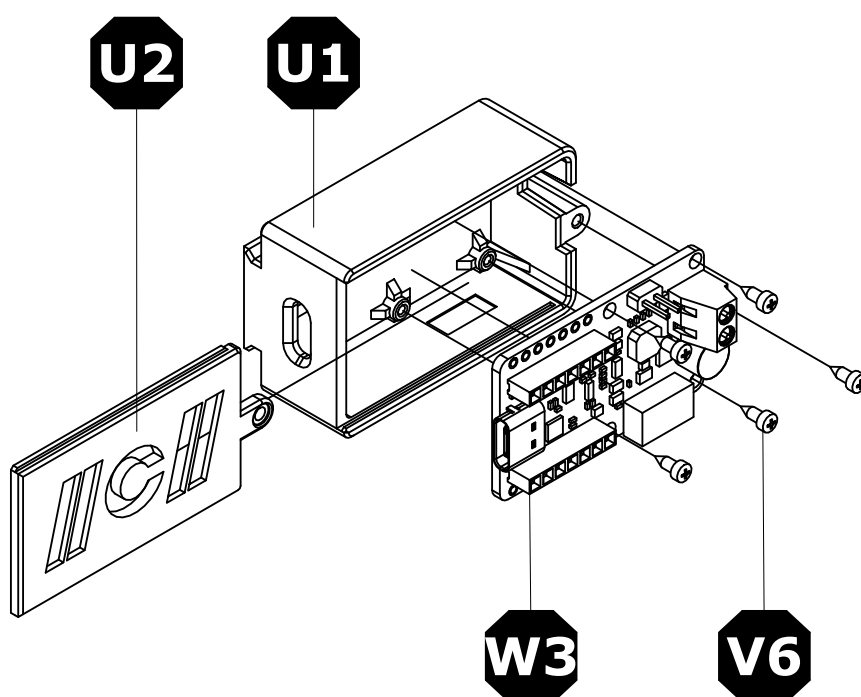
ASSEMBLE THE
ENCLOSURE
AROUND THE
SERVO DRIVER
BOARD.

ENCLOSURE	
T1	1x
ENCLOSURE	
T2	1x
SERVO DRIVER BOARD	
W2	1x
SELF-TAPPING SCREW M2X6	
V6	5x

STEP 2

CHOOSE ENCLOSURE DEPENDS ON YOUR SERVO DRIVE BOARD

ASSEMBLE THE
ENCLOSURE
AROUND THE
SERVO DRIVER
BOARD.



ENCLOSURE	
U1	1x
ENCLOSURE	
U2	1x
SERVO DRIVER BOARD	
W3	1x
SELF-TAPPING SCREW M2X6	
V6	5x

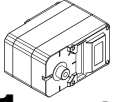
OPTIONAL

ELROBOT

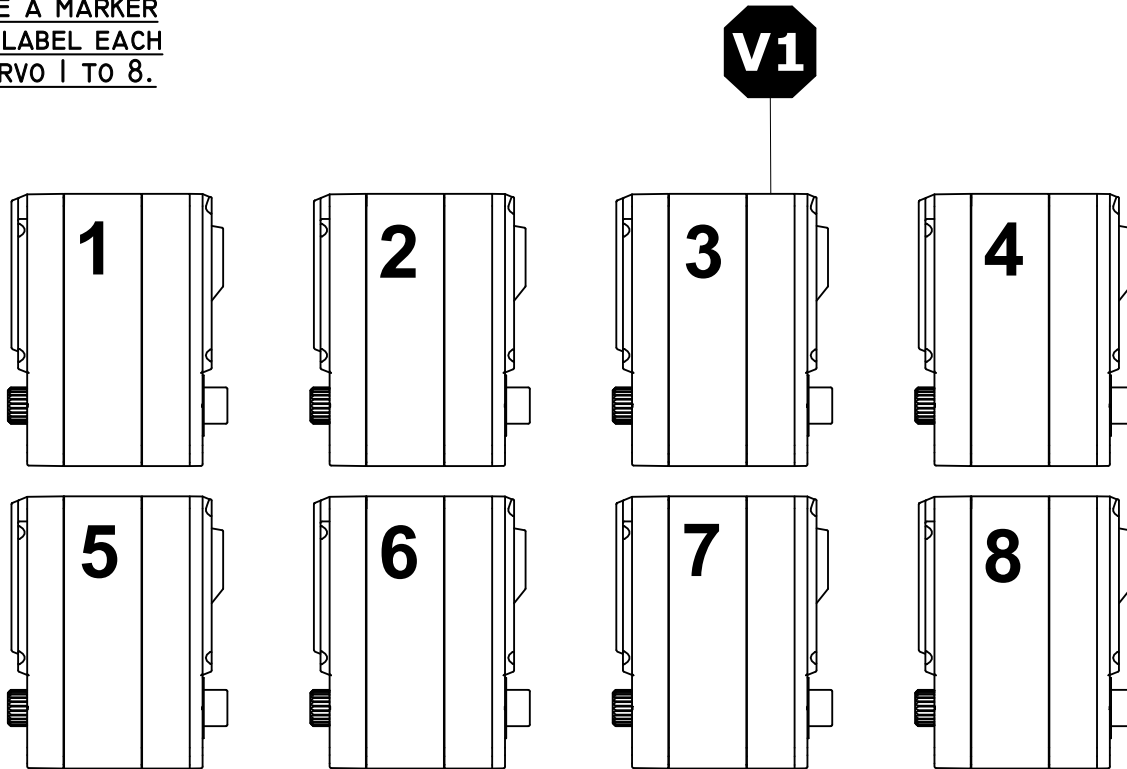
STEP 3

USE A MARKER
TO LABEL EACH
SERVO 1 TO 8.

SERVO ST3215



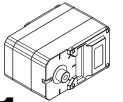
V1 8x



STEP 4

SET SERVO IDS USING
APP.
! REPEAT 8 TIMES !

SERVO ST3215



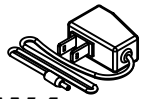
V1 8x

SERVO WIRES

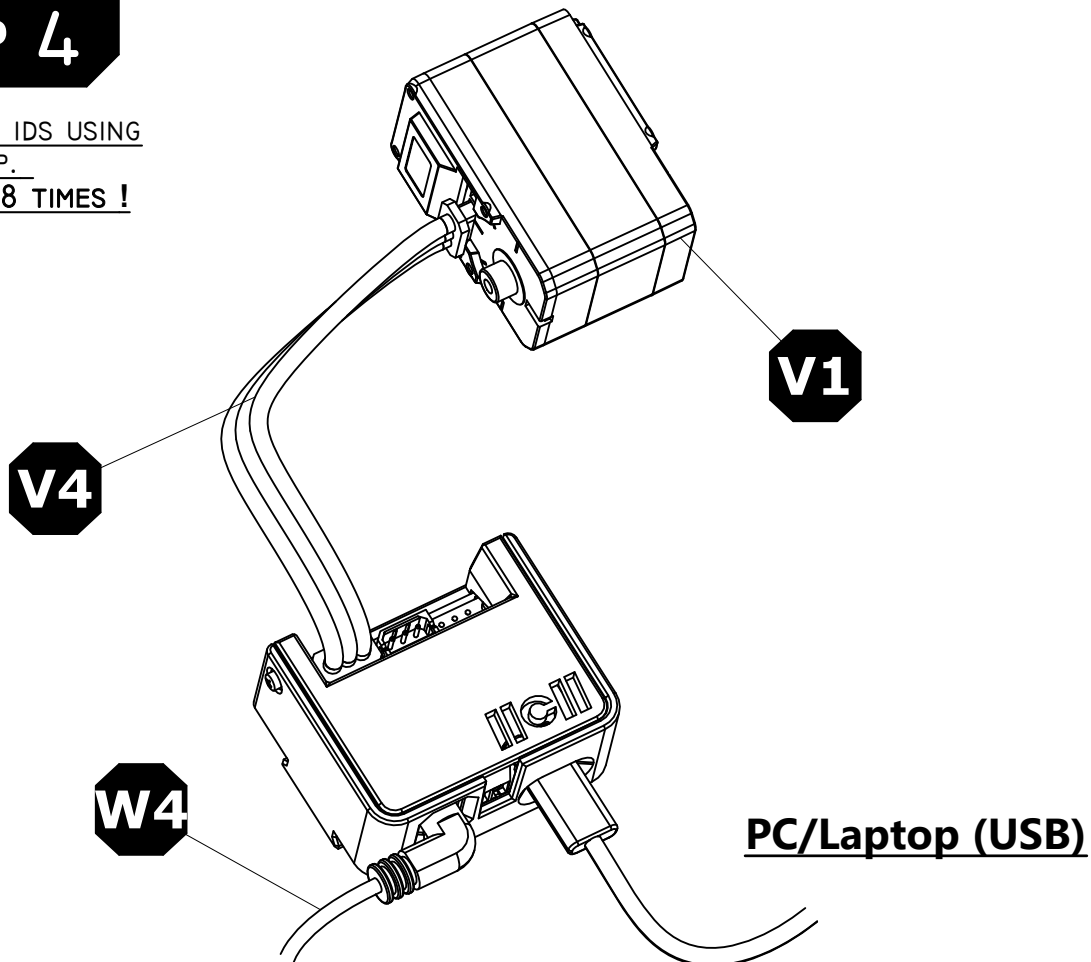


V4 1x

POWER SUPPLY



W4 1x

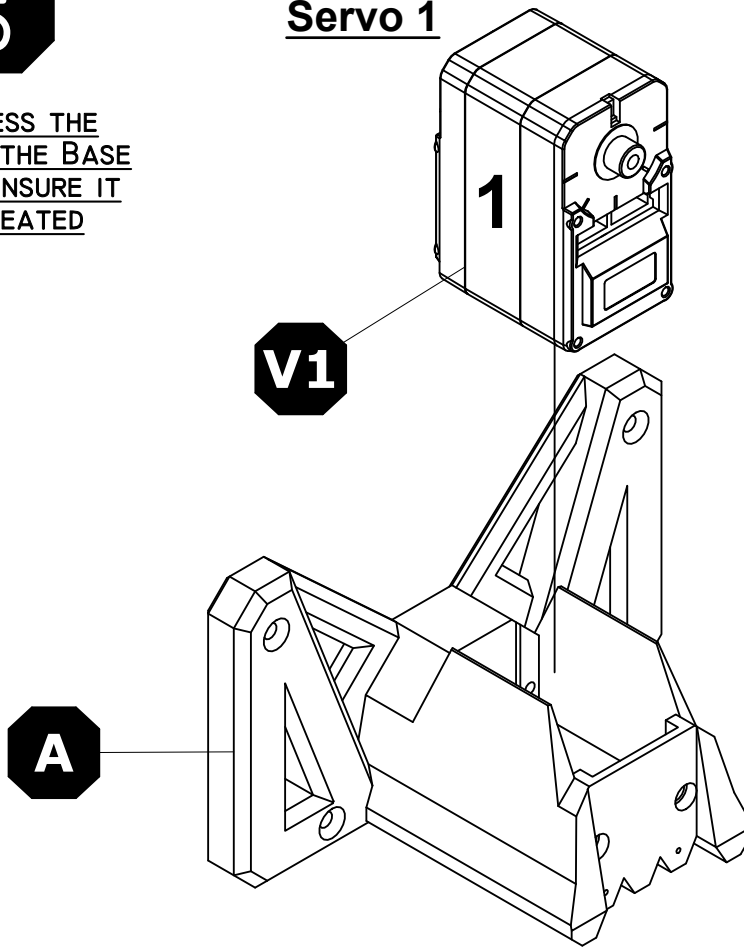


ELROBOT

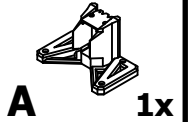
STEP 5

GENTLY PRESS THE
SERVO 1 INTO THE BASE
FOOTPRINT. ENSURE IT
IS FULLY SEATED

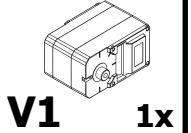
Servo 1



BASE



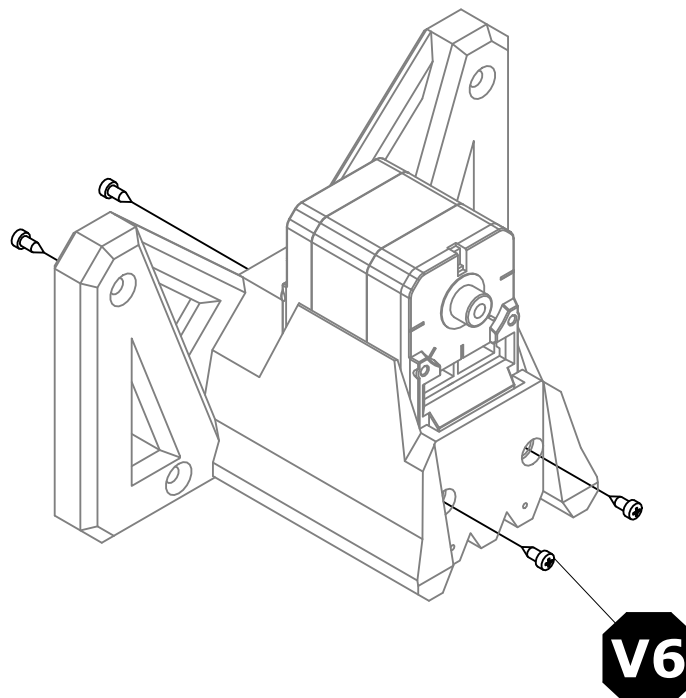
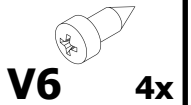
SERVO ST3215



STEP 6

ATTACH THE SERVO
1 TO THE BASE

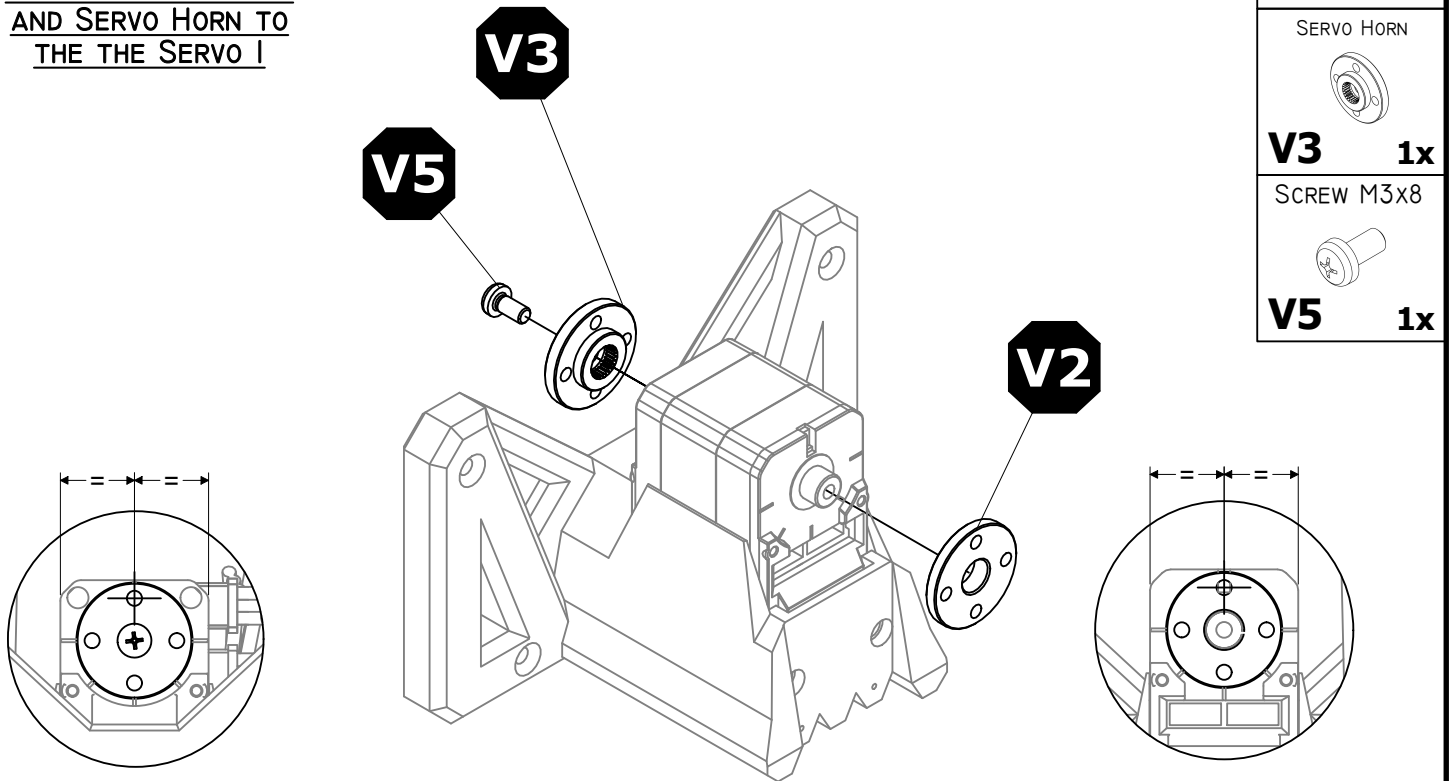
SELF-TAPPING
SCREW M2x6



ELROBOT

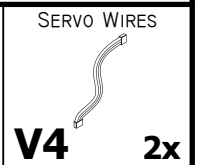
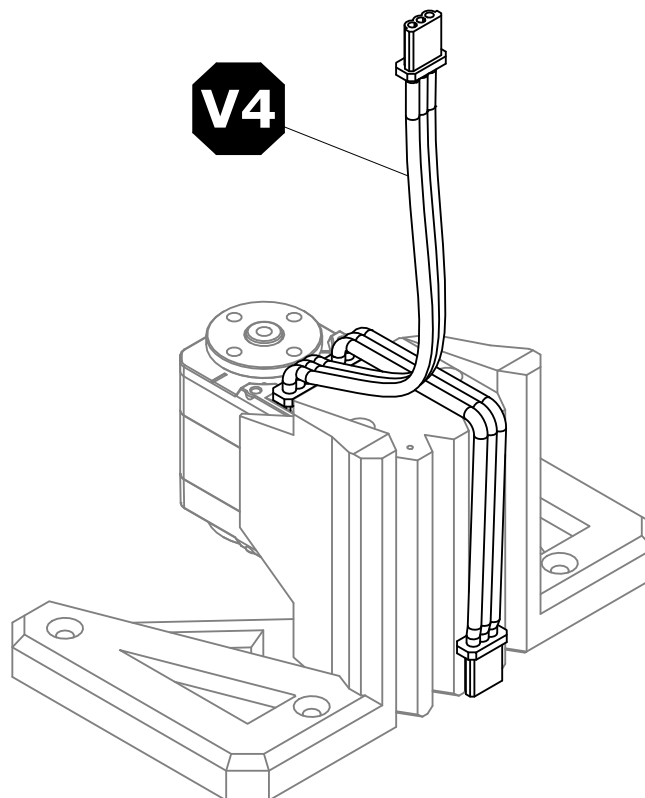
STEP 8

ATTACH SERVO IDLER
AND SERVO HORN TO
THE THE SERVO I



STEP 9

CONNECT AND ROUTE
THE SERVO WIRES AS
SHOWN.



ELROBOT

STEP 10

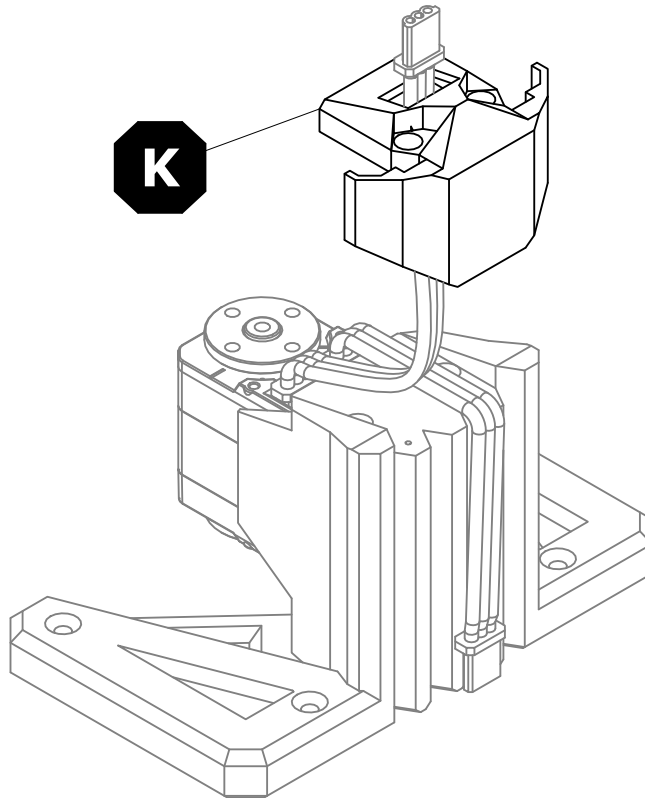
ROUTE THE SERVO
WIRES THROUGH THE
SHIELD AS SHOWN.

BASE Shield



K

1x



STEP 11

ATTACH THE JOINT 1
TO THE SERVO 1

JOINT 1



B

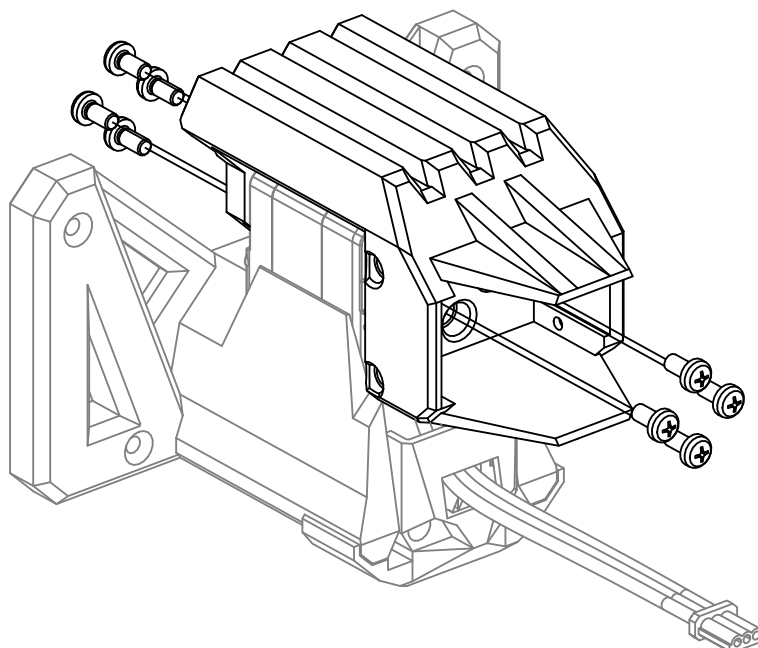
1x

SCREW M3x8



V5

8x



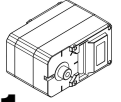
ELROBOT

STEP 12

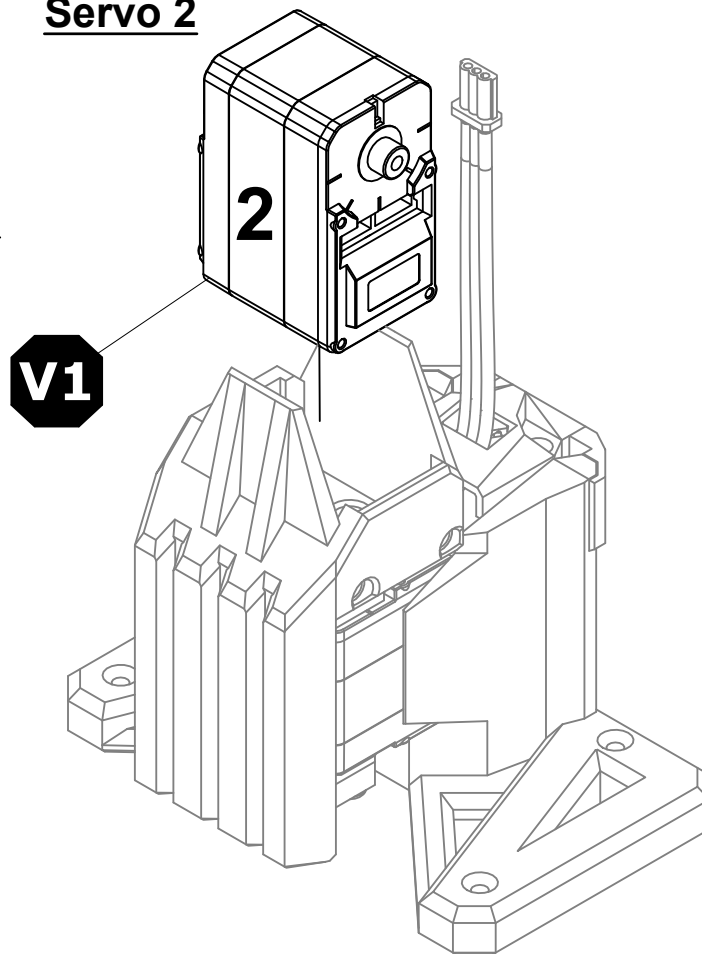
Servo 2

GENTLY PRESS THE
SERVO 2 INTO THE
JOINT 1 FOOTPRINT.
ENSURE IT IS FULLY
SEATED

SERVO ST3215



V1 1x



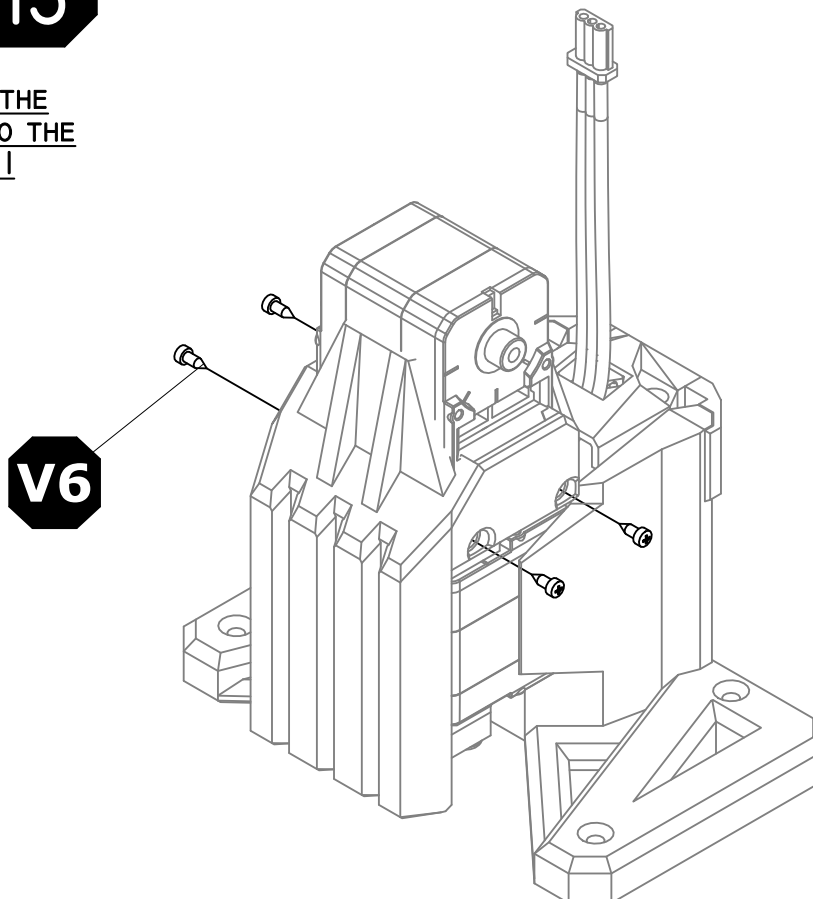
STEP 13

ATTACH THE
SERVO 2 TO THE
JOINT 1

SELF-TAPPING
SCREW M2x6



V6 4x



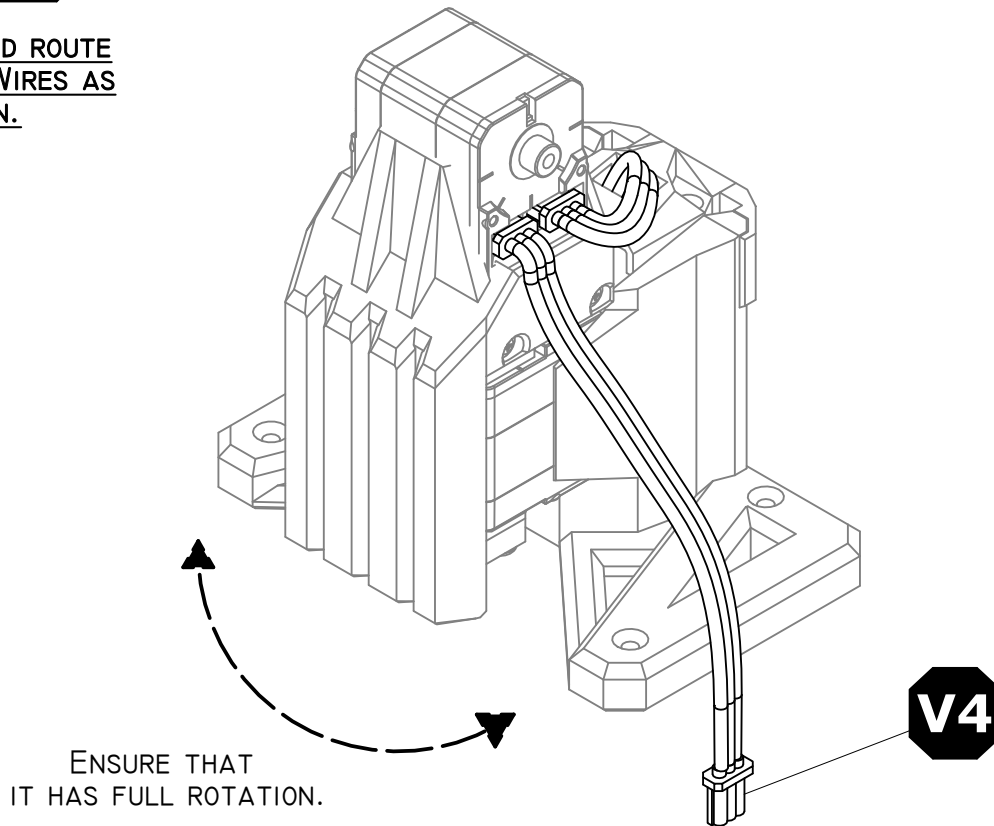
ELROBOT

STEP 14

CONNECT AND ROUTE
THE SERVO WIRES AS
SHOWN.

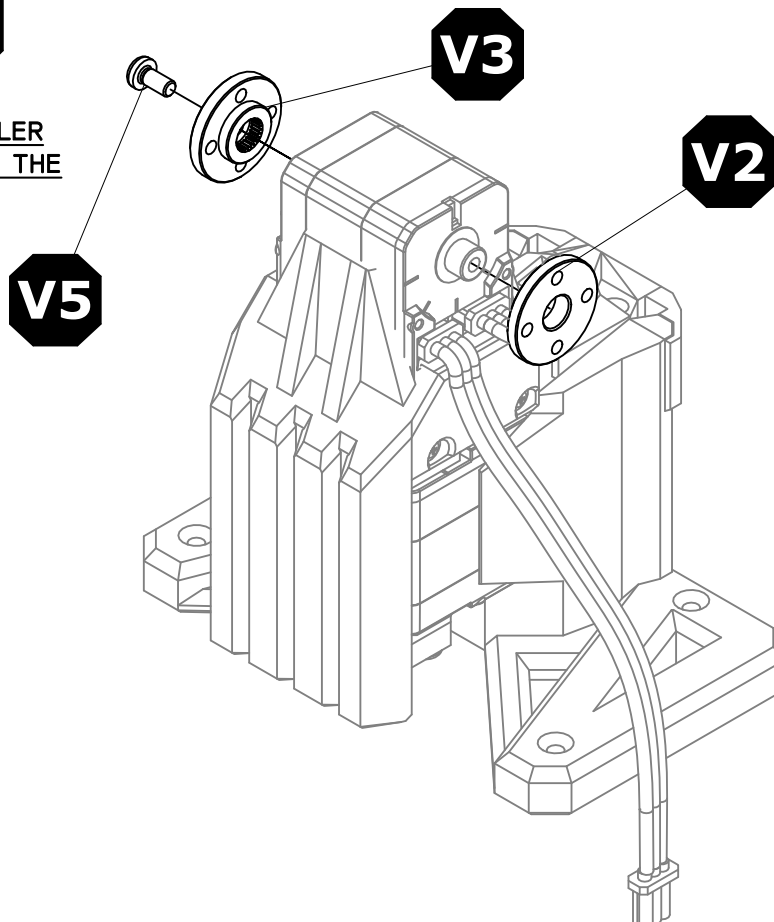
SERVO WIRES

V4 1x



STEP 15

ATTACH SERVO IDLER
AND SERVO HORN TO THE
SERVO 2



IDLER HORN

V2 1x

SERVO HORN

V3 1x

SCREW M3x8

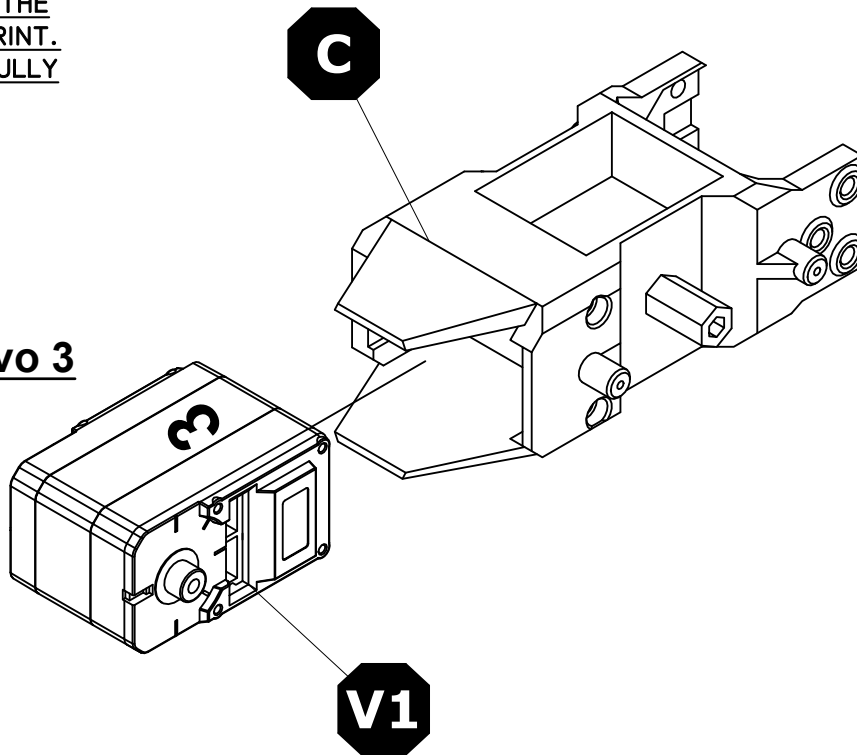
V5 1x

ELROBOT

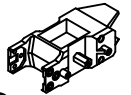
STEP 16

GENTLY PRESS THE
SERVO 3 INTO THE
JOINT 2 FOOTPRINT.
ENSURE IT IS FULLY
SEATED

Servo 3

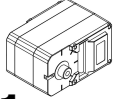


JOINT 2



C 1x

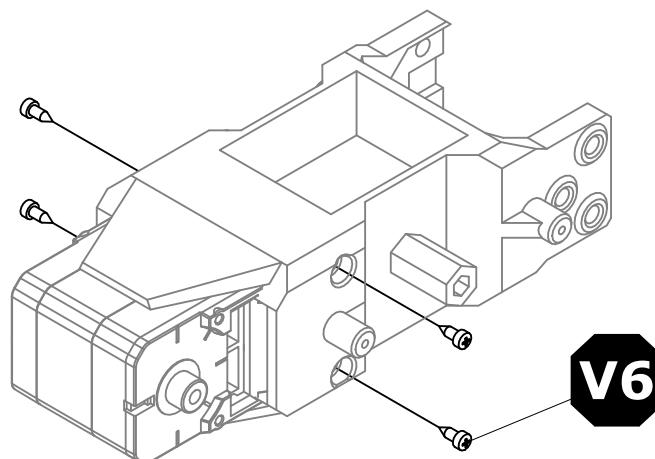
SERVO ST3215



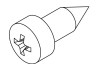
V1 1x

STEP 17

ATTACH THE
SERVO 3 TO THE
JOINT 2



SELF-TAPPING
SCREW M2X6



V6 4x

ELROBOT

STEP 18

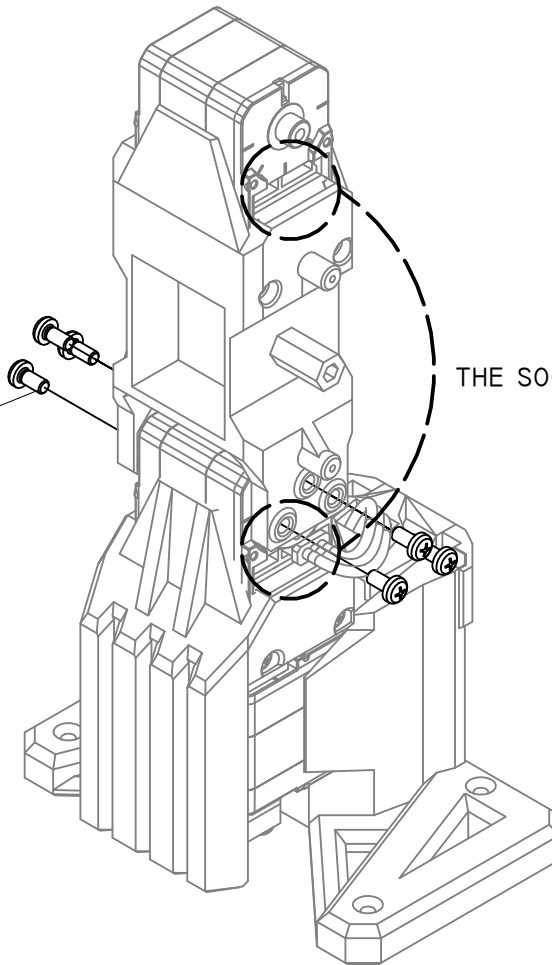
ATTACH THE JOINT 2
TO THE SERVO 2

SCREW M3x8



V5 6x

V5



ENSURE THAT
THE SOCKETS ARE ON ONE SIDE.

STEP 19

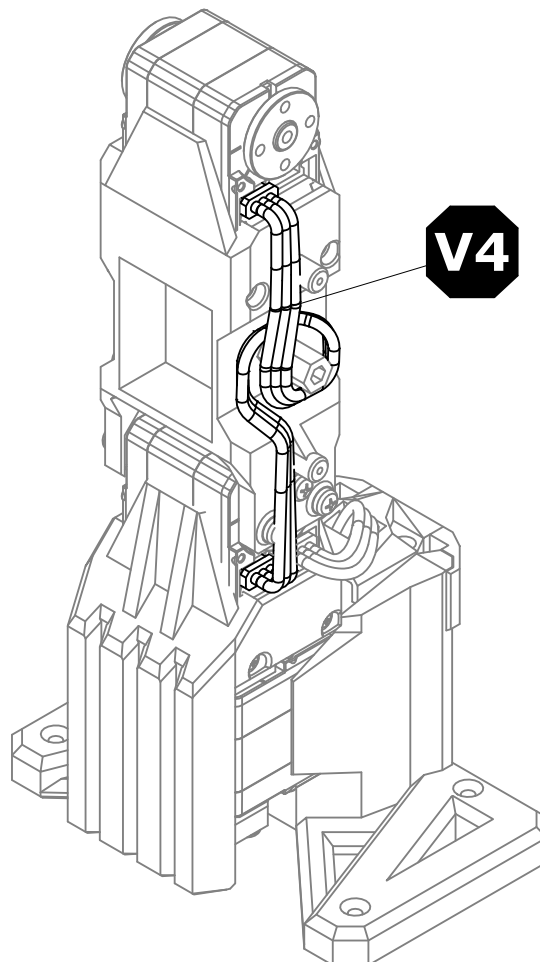
CONNECT AND ROUTE
THE SERVO WIRES AS
SHOWN.

SERVO WIRES



V4 1x

V4



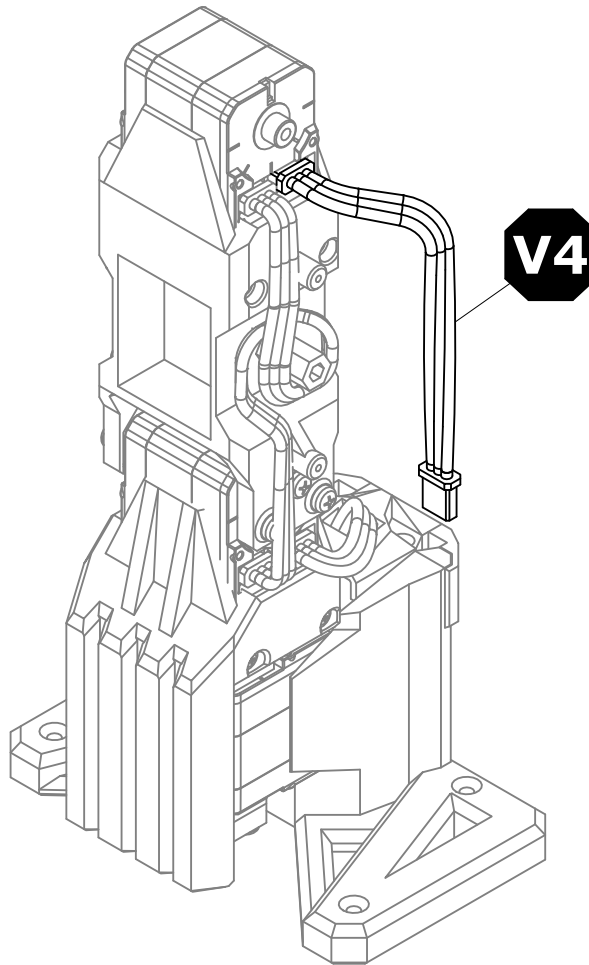
ELROBOT

STEP 20

CONNECT THE SERVO
WIRES AS SHOWN.

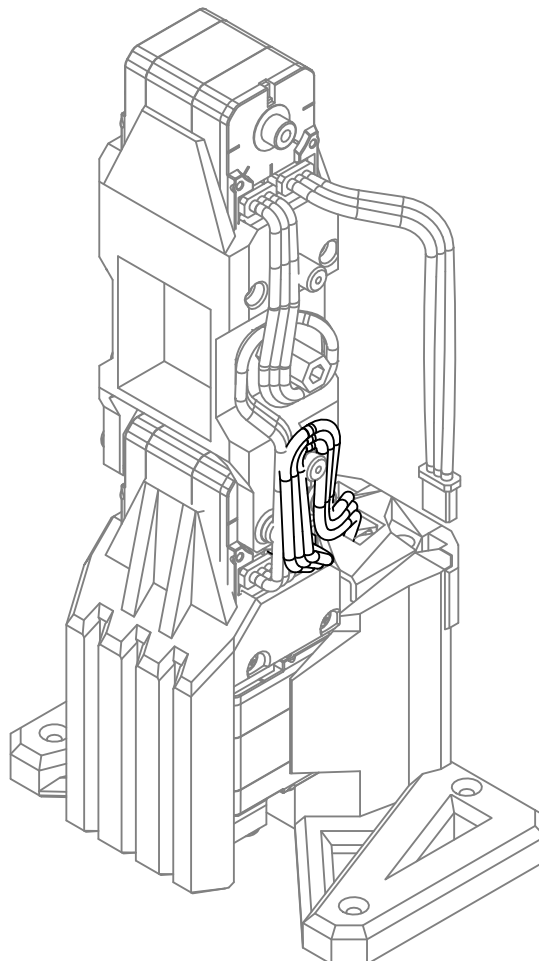
SERVO WIRES

V4 1x



STEP 21

ROUTE THE SERVO
WIRES AS SHOWN.

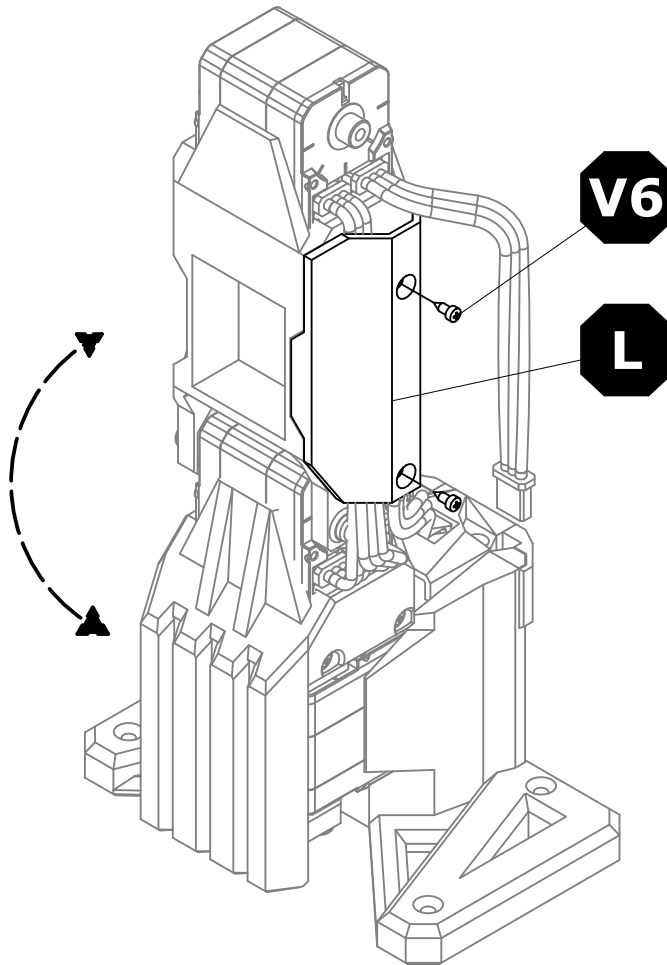


ELROBOT

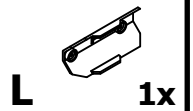
STEP 22

ATTACH THE SHIELD
TO THE JOINT 2

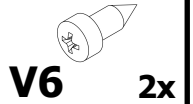
ENSURE THAT
IT HAS FULL ROTATION.



JOINT 2 SHIELD

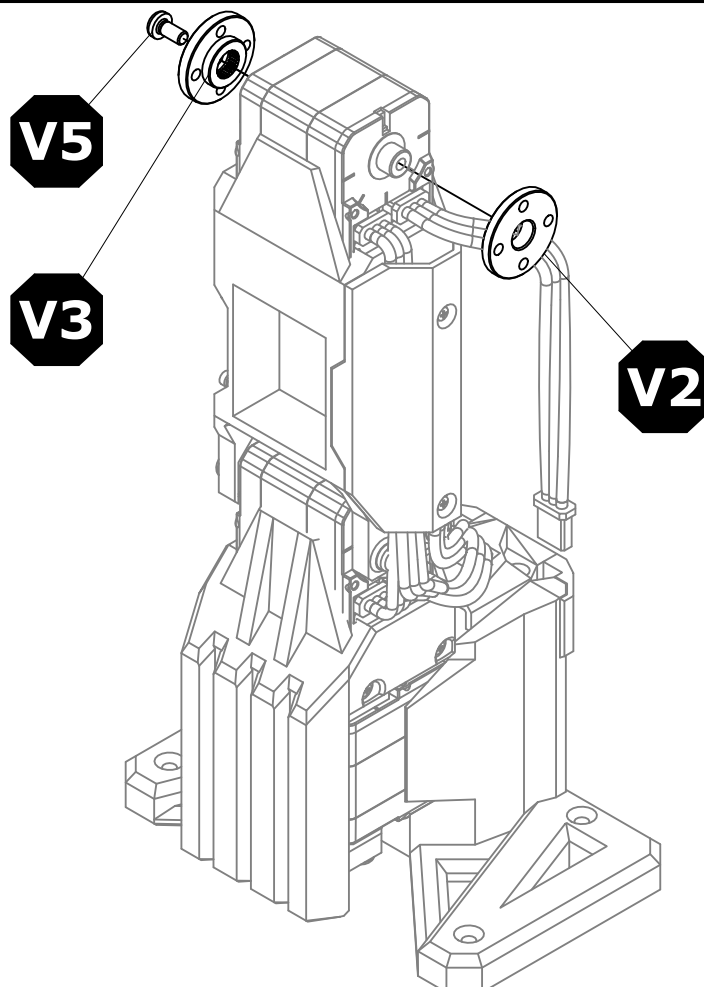


SELF-TAPPING
SCREW M2x6

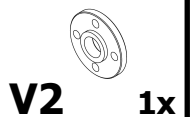


STEP 23

ATTACH SERVO IDLER
AND SERVO HORN TO
THE THE SERVO 3



IDLER HORN



SERVO HORN



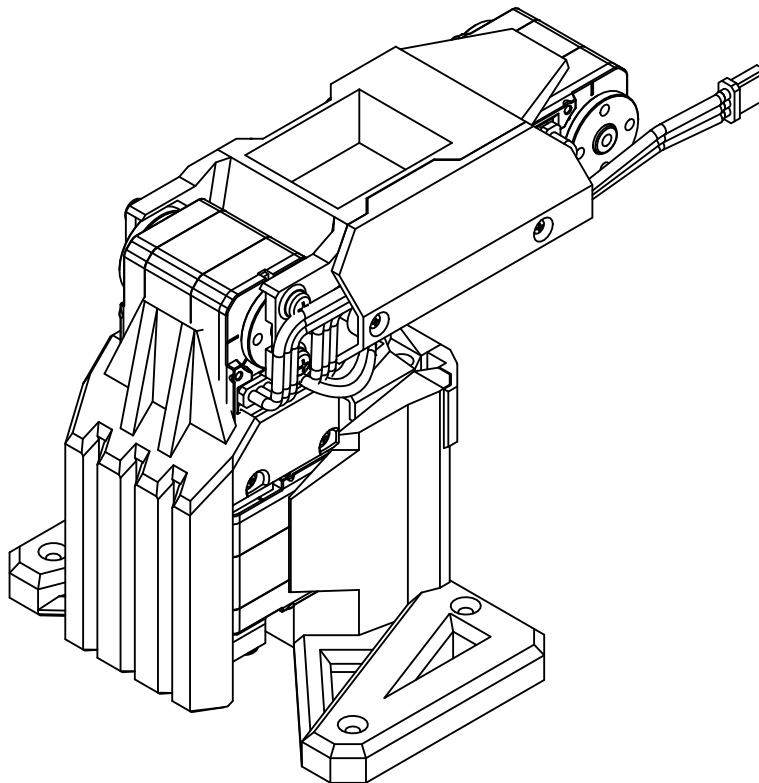
SCREW M3x8



ELROBOT

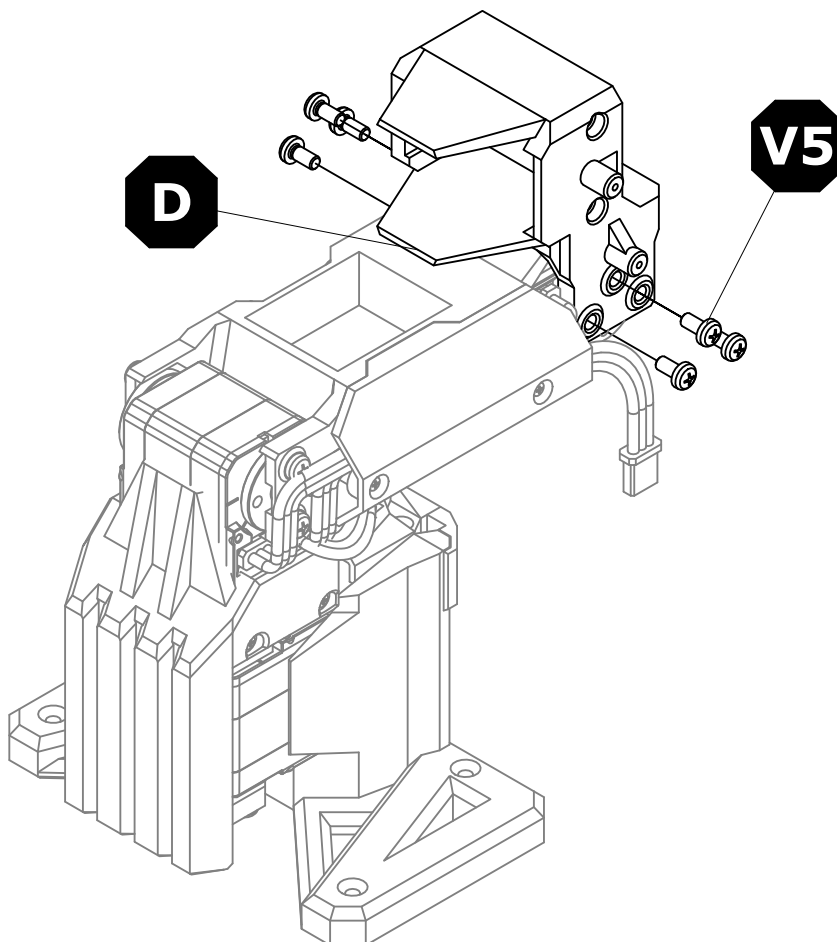
STEP 24

TURN ELROBOT
LIKE SHOWN

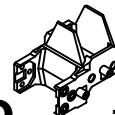


STEP 25

ATTACH THE JOINT 3
TO THE SERVO 3



JOINT 3



D 1x

SCREW M3x8



V5 6x

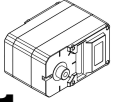
ELROBOT

STEP 26

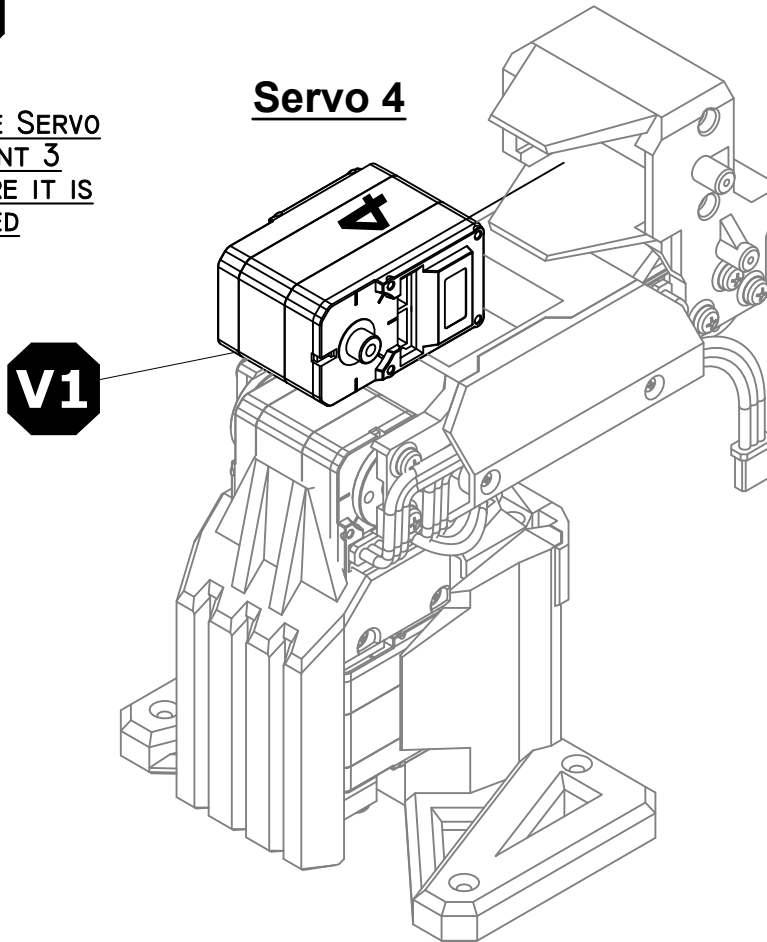
GENTLY PRESS THE SERVO 4 INTO THE JOINT 3 FOOTPRINT. ENSURE IT IS FULLY SEATED

Servo 4

SERVO ST3215



V1 1x



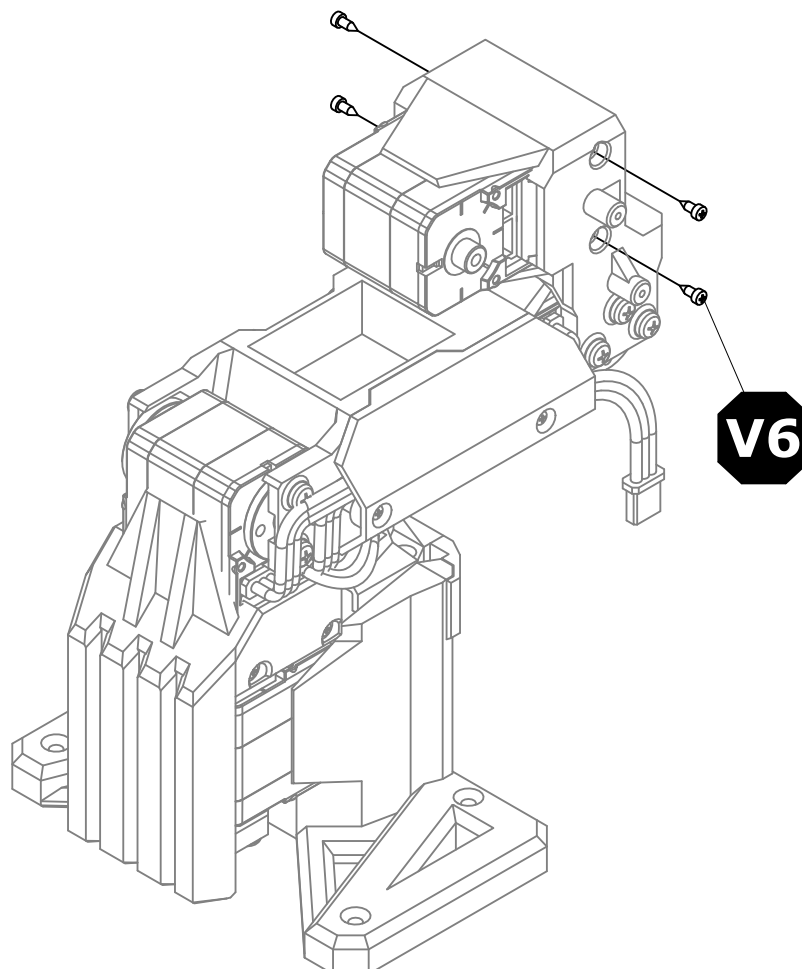
STEP 27

ATTACH THE SERVO 4 TO THE JOINT 3

SELF-TAPPING SCREW M2x6



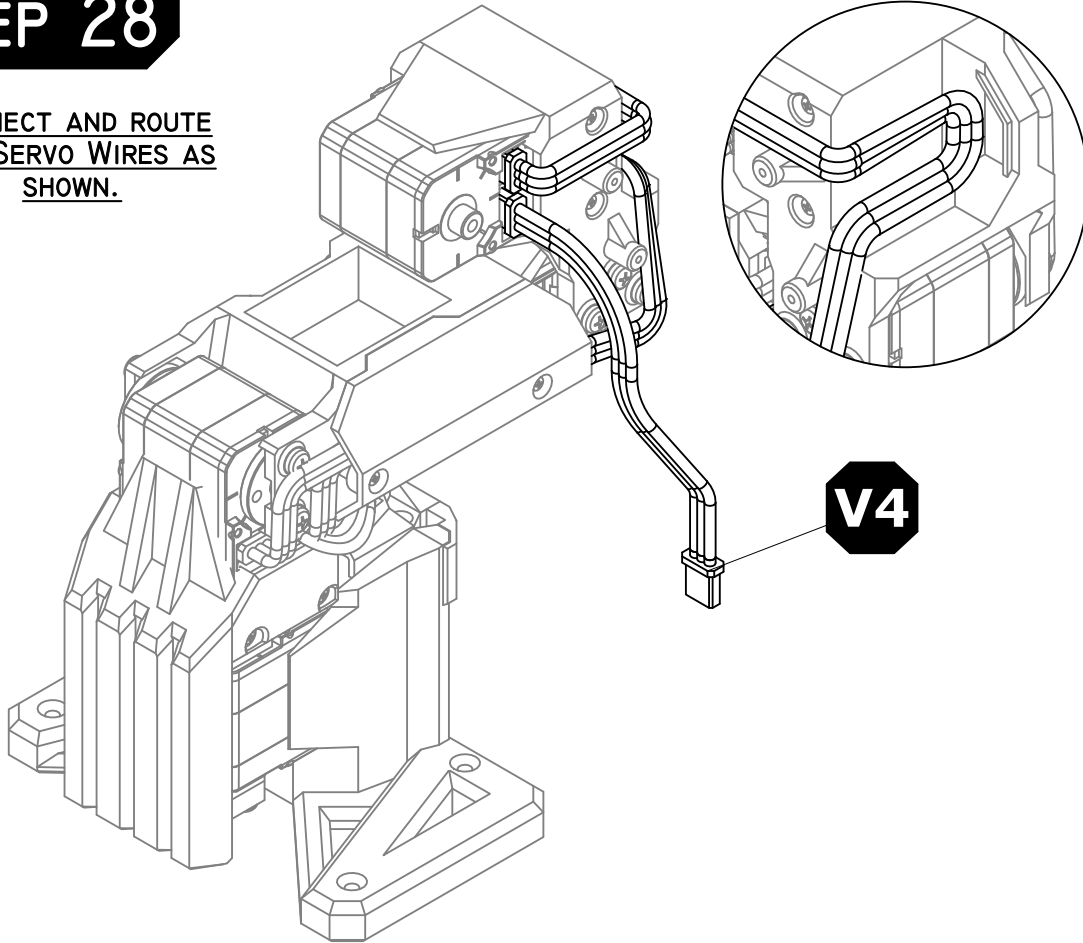
V6 4x



ELROBOT

STEP 28

CONNECT AND ROUTE
THE SERVO WIRES AS
SHOWN.



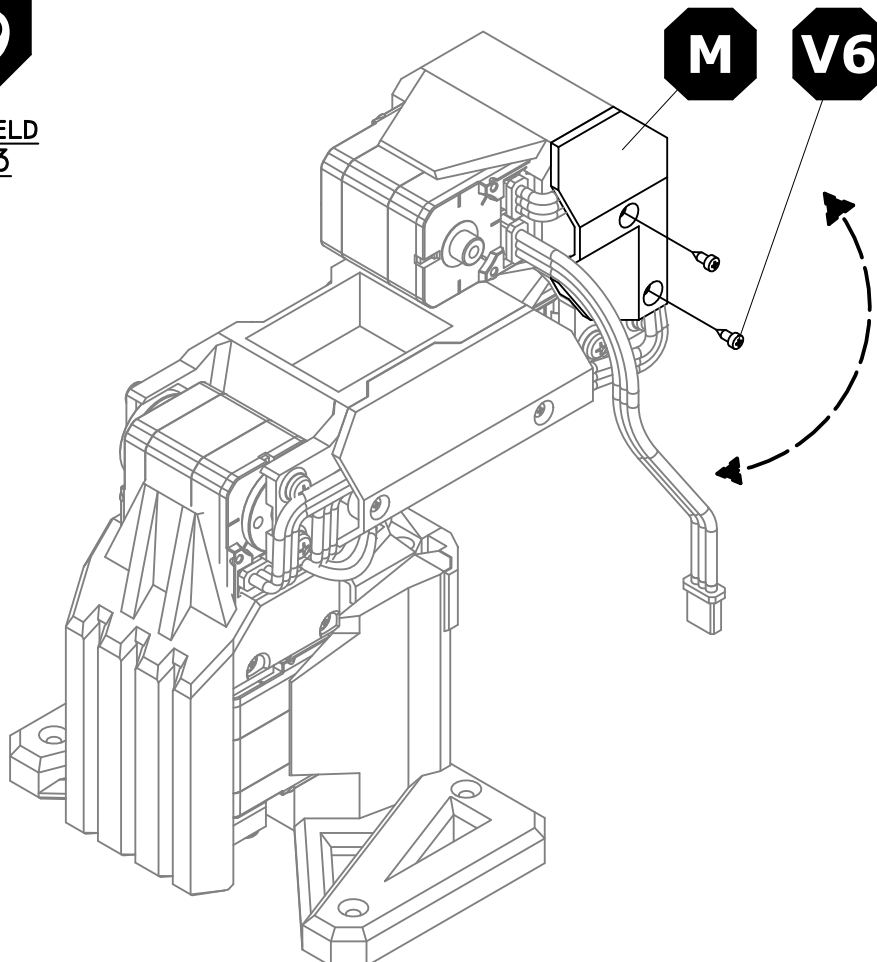
SERVO WIRES

V4 1x

V4

STEP 29

ATTACH THE SHIELD
TO THE JOINT 3



JOINT 3 SHIELD
DRIVE SIDE

M 1x

SELF-TAPPING
SCREW M2x6

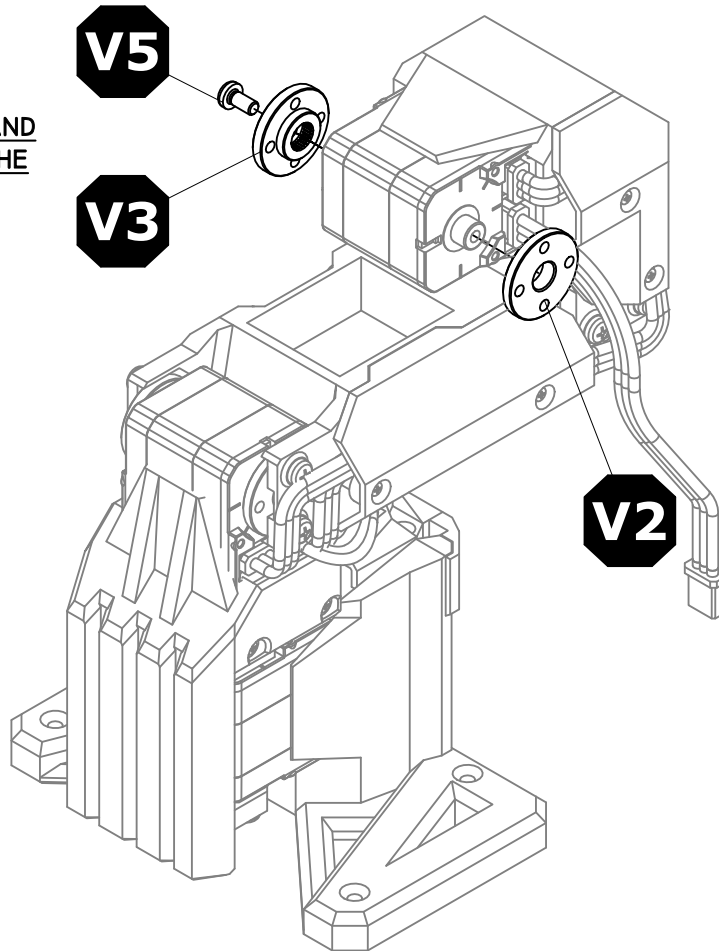
V6 2x

ENSURE THAT
IT HAS FULL ROTATION.

ELROBOT

STEP 30

ATTACH SERVO IDLER AND
SERVO HORN TO THE THE
SERVO 4



IDLER HORN



V2 1x

SERVO HORN



V3 1x

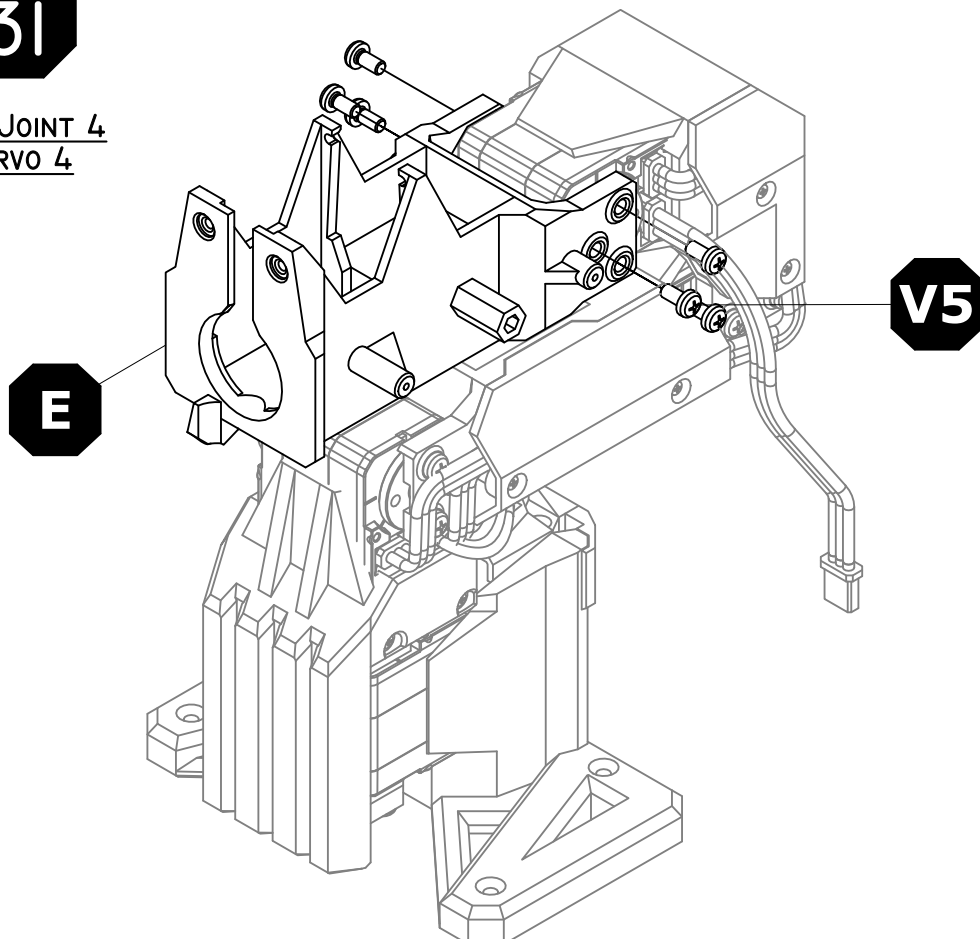
SCREW M3x8



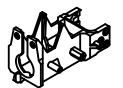
V5 1x

STEP 31

ATTACH THE JOINT 4
TO THE SERVO 4



JOINT 4



E 1x

SCREW M3x8

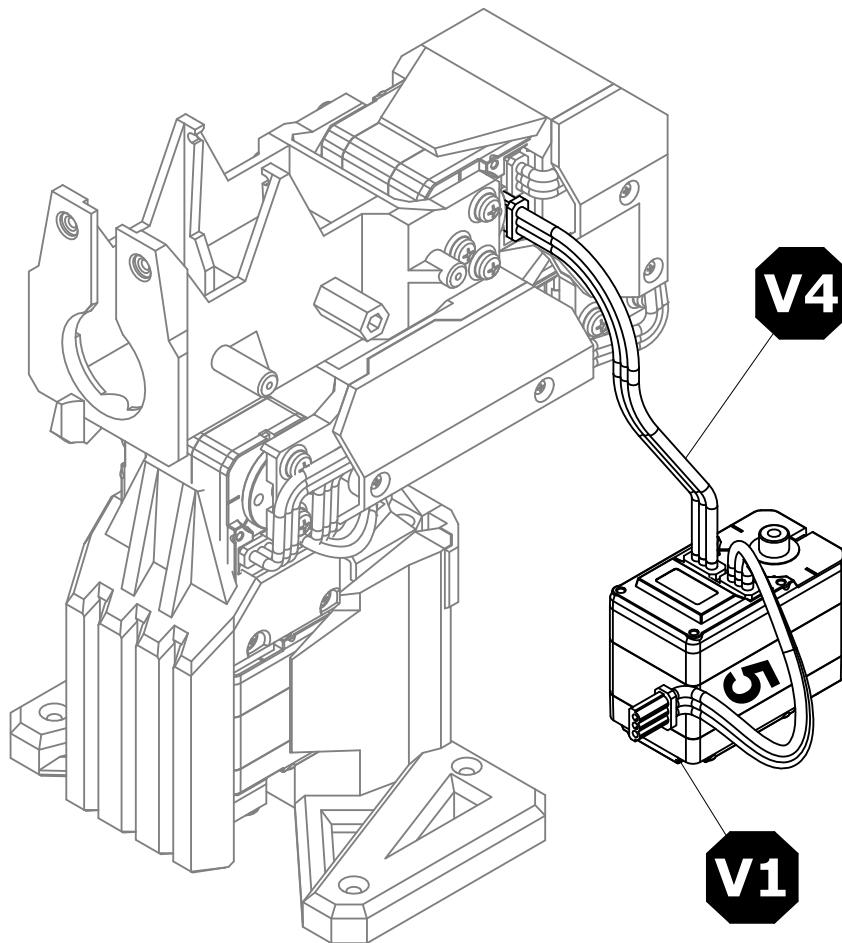


V5 6x

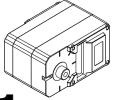
ELROBOT

STEP 32

CONNECT AND ROUTE
THE SERVO WIRES AS
SHOWN.



SERVO ST3215



V1 1x

SERVO WIRES



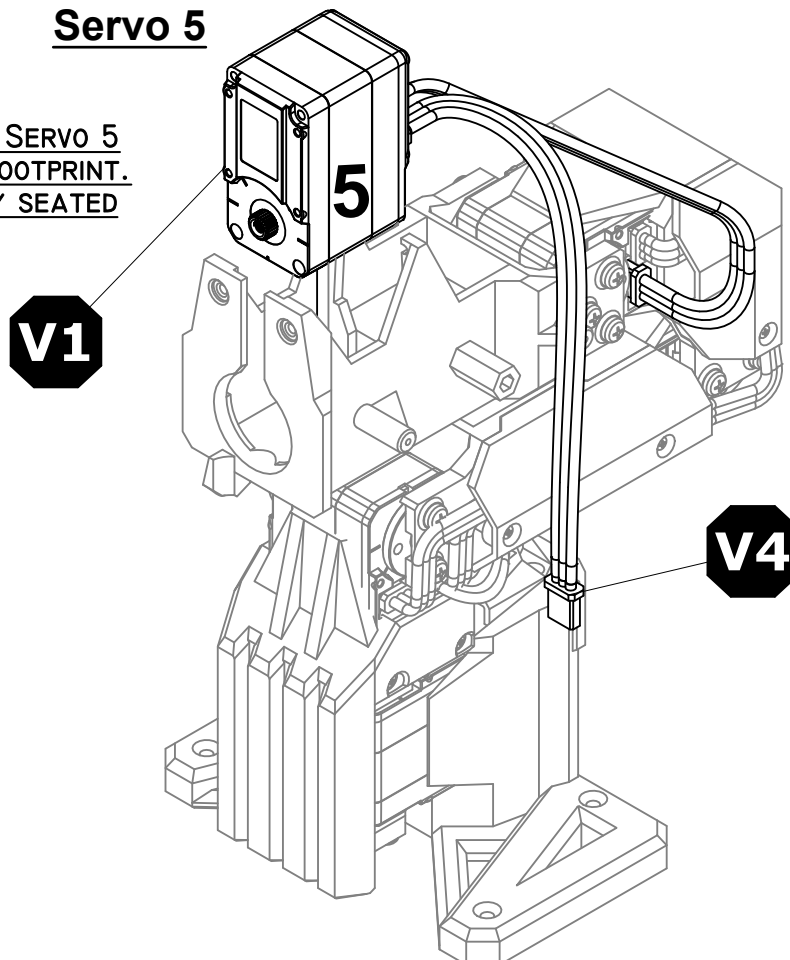
V4 1x

Servo 5

STEP 33

Servo 5

GENTLY PRESS THE SERVO 5
INTO THE JOINT 4 FOOTPRINT.
ENSURE IT IS FULLY SEATED



ELROBOT

STEP 34

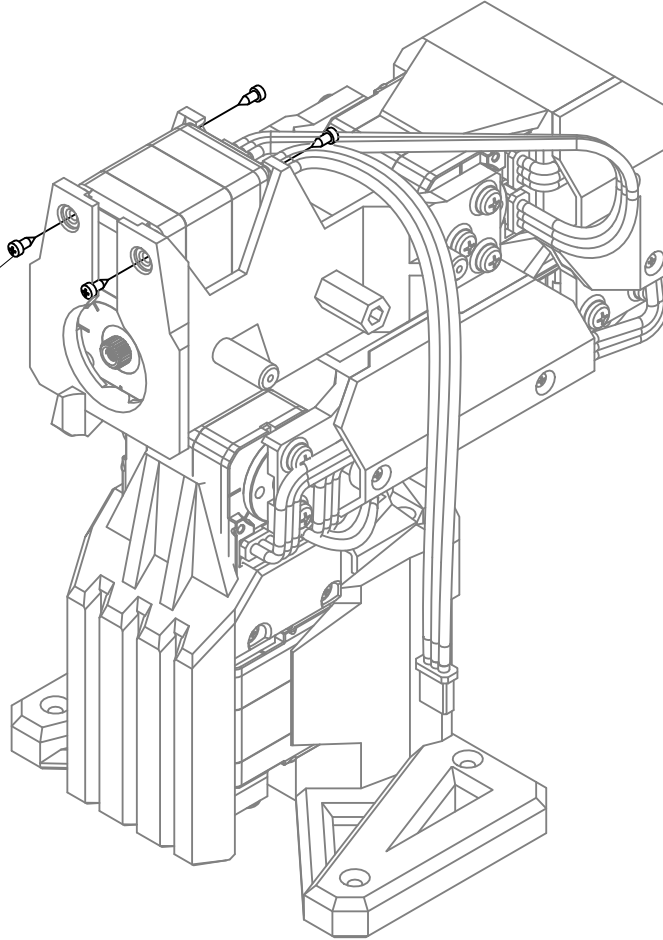
ATTACH THE
SERVO 5 TO THE
JOINT 4

SELF-TAPPING
SCREW M2X6



V6 4x

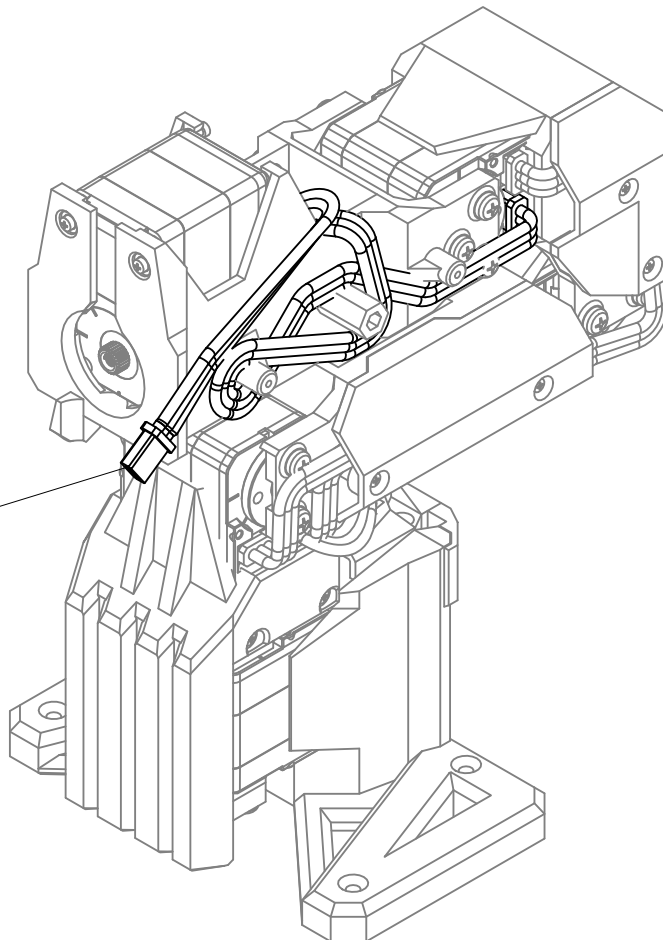
V6



STEP 35

ROUTE THE SERVO
WIRES AS SHOWN.

V4

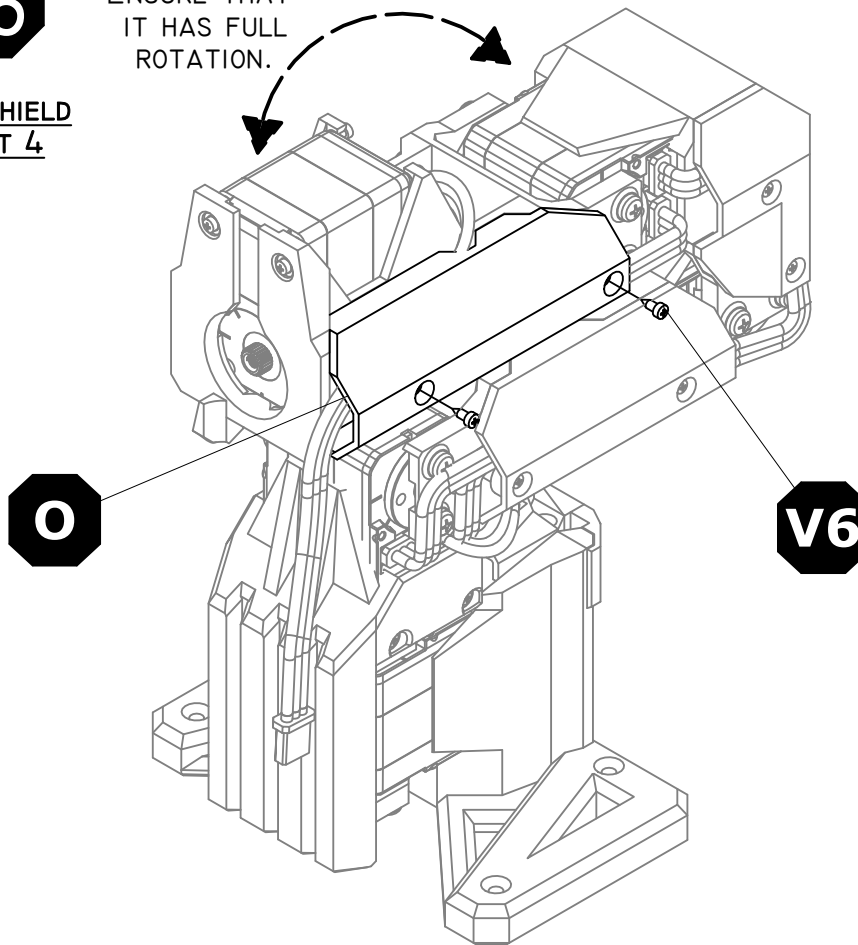


ELROBOT

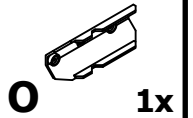
STEP 36

ATTACH THE SHIELD
TO THE JOINT 4

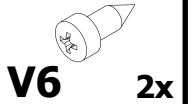
ENSURE THAT
IT HAS FULL
ROTATION.



JOINT 4 SHIELD

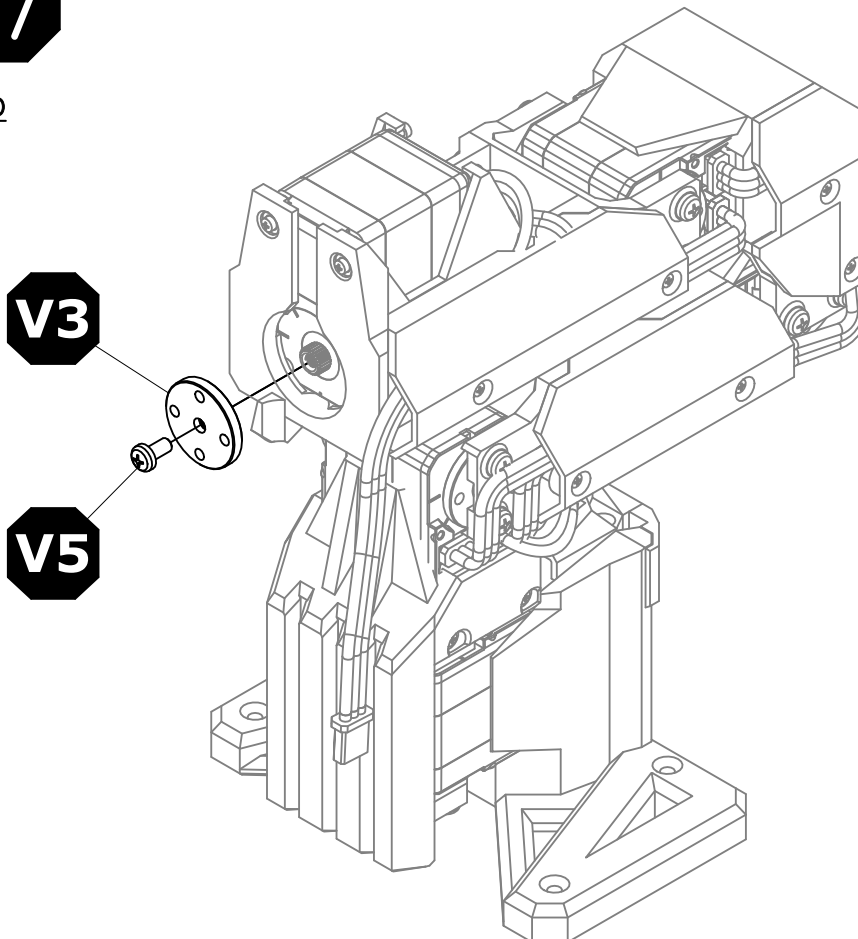


SELF-TAPPING
SCREW M2x6

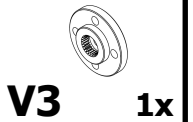


STEP 37

ATTACH SERVO
HORN TO THE
SERVO 5



SERVO HORN



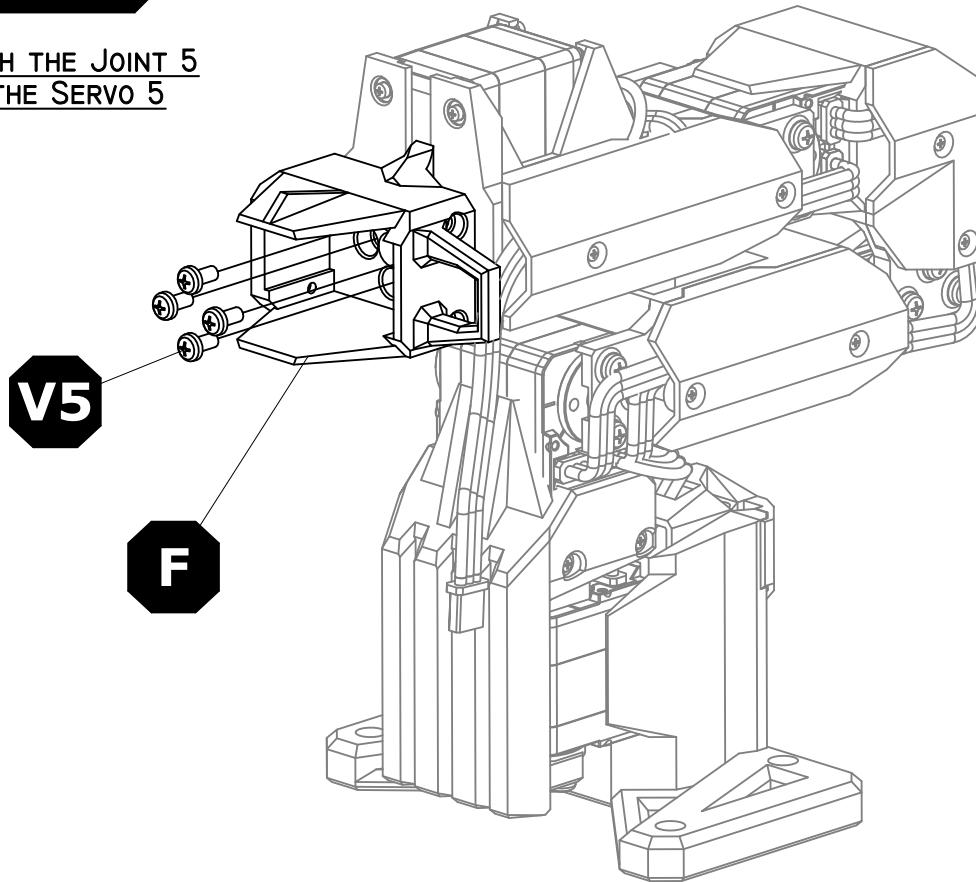
SCREW M3x8



ELROBOT

STEP 38

ATTACH THE JOINT 5
TO THE SERVO 5



JOINT 5



F **1x**

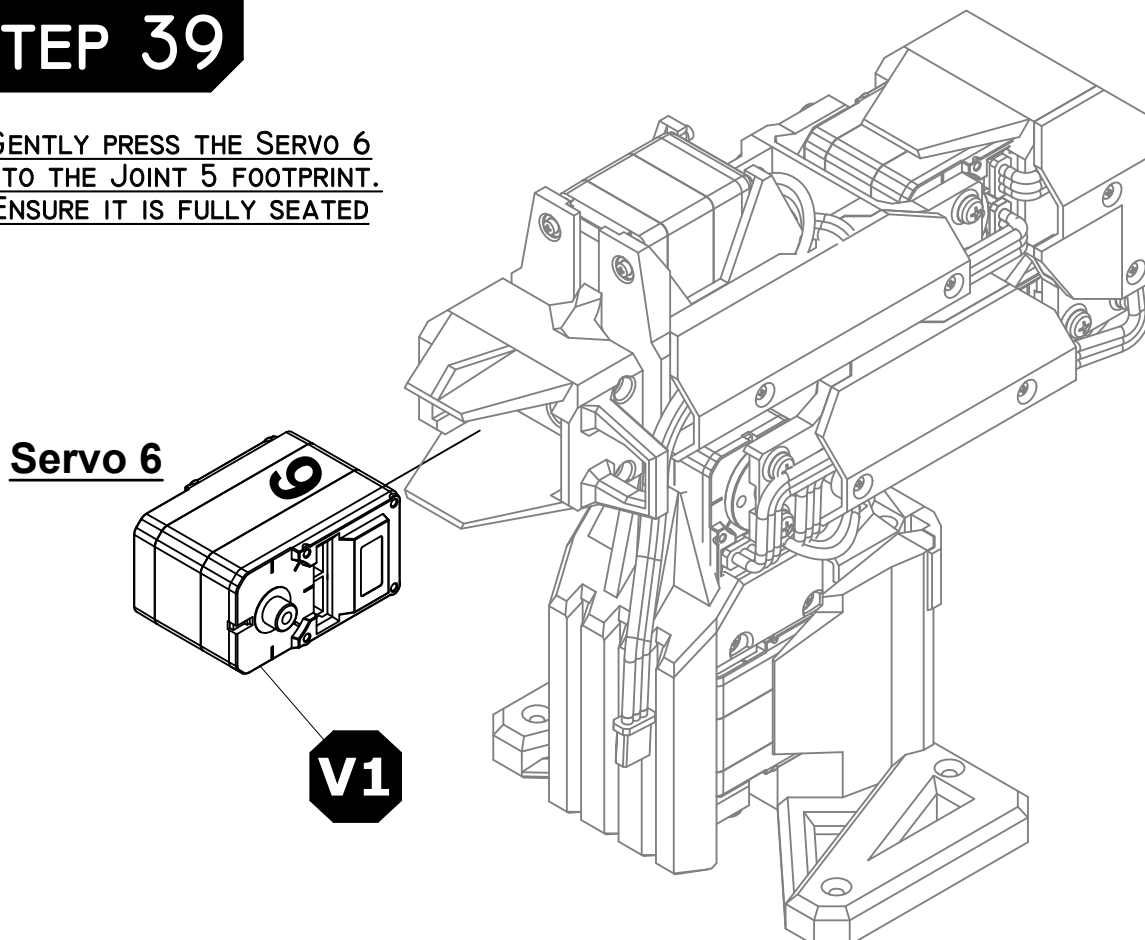
SCREW M3x8



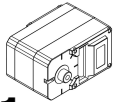
V5 **4x**

STEP 39

GENTLY PRESS THE SERVO 6
INTO THE JOINT 5 FOOTPRINT.
ENSURE IT IS FULLY SEATED



SERVO ST3215



V1 **1x**

ELROBOT

STEP 40

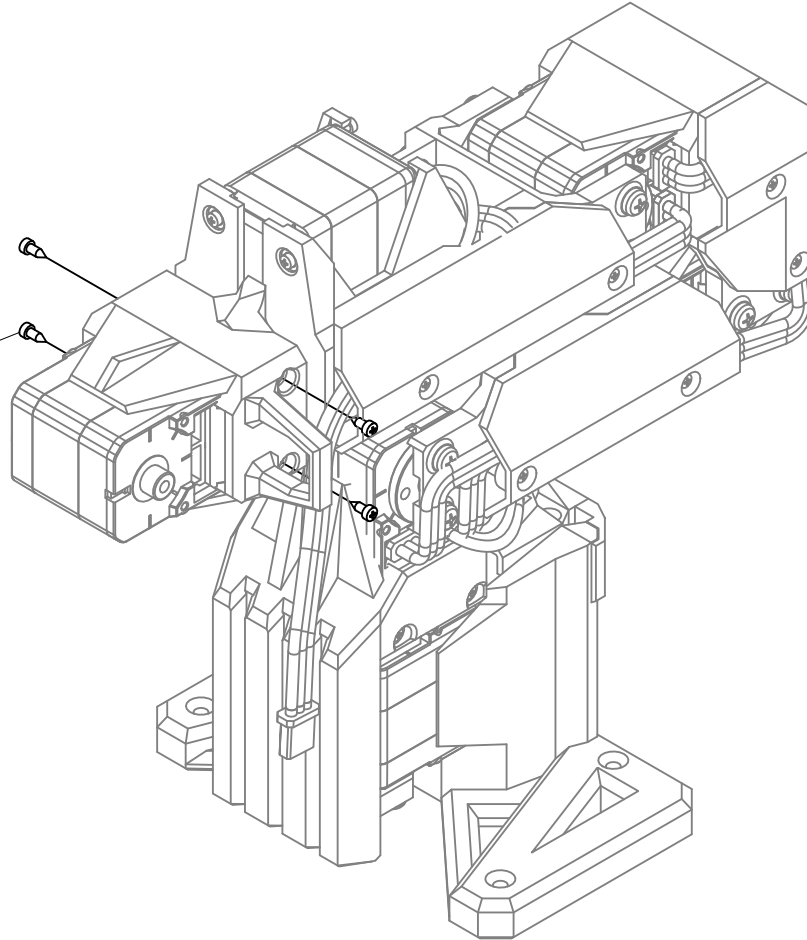
ATTACH THE
SERVO 6 TO THE
JOINT 5

SELF-TAPPING
SCREW M2X6



V6 4x

V6



STEP 41

CONNECT AND ROUTE
THE SERVO WIRES AS
SHOWN.

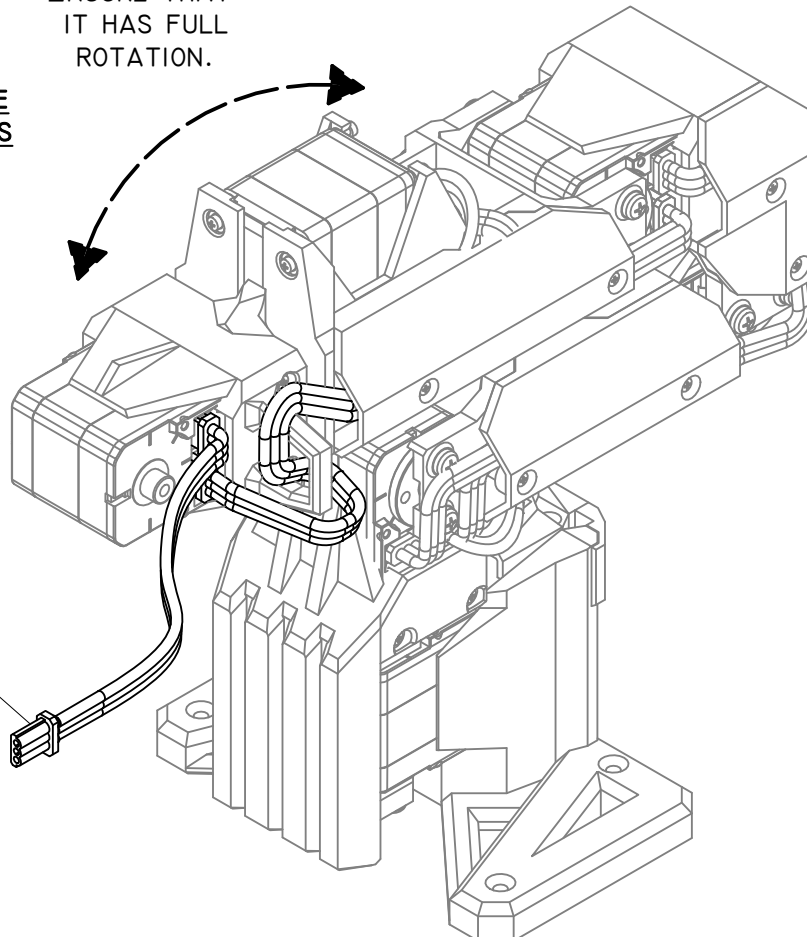
ENSURE THAT
IT HAS FULL
ROTATION.

SERVO WIRES



V4 1x

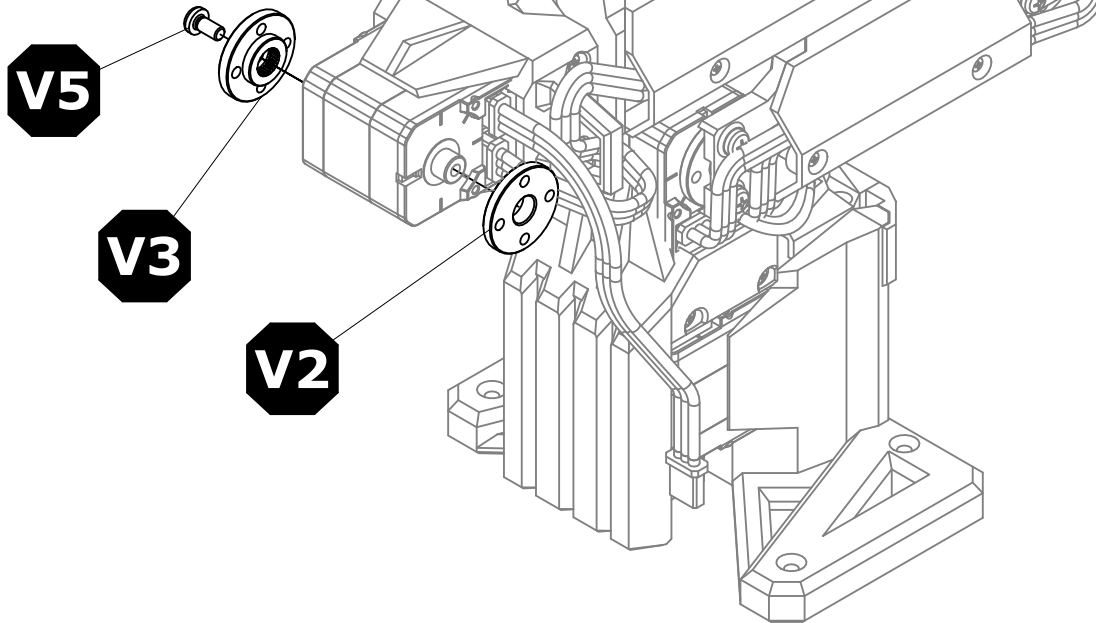
V4



ELROBOT

STEP 42

ATTACH SERVO IDLER AND
SERVO HORN TO THE THE
SERVO 6



IDLER HORN



V2 1x

SERVO HORN



V3 1x

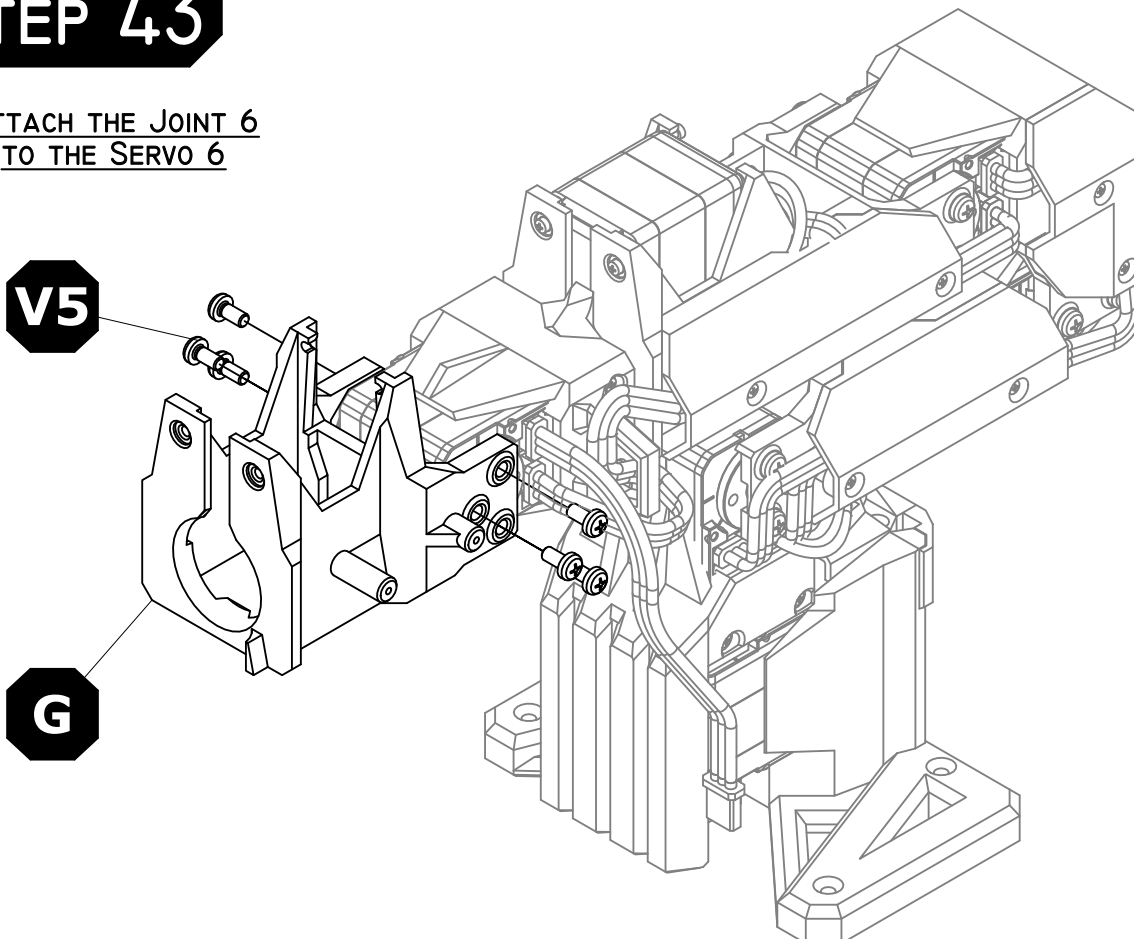
SCREW M3x8



V5 1x

STEP 43

ATTACH THE JOINT 6
TO THE SERVO 6



JOINT 6



G 1x

SCREW M3x8

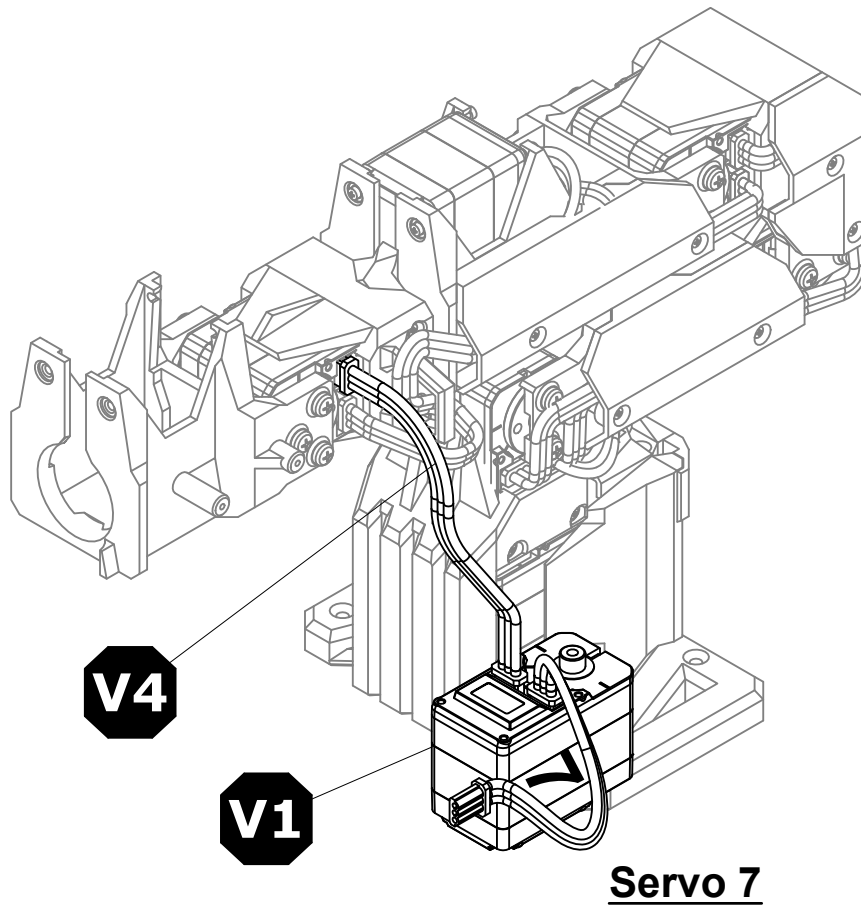


V5 6x

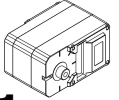
ELROBOT

STEP 44

CONNECT AND ROUTE
THE SERVO WIRES AS
SHOWN.



SERVO ST3215



V1 **1x**

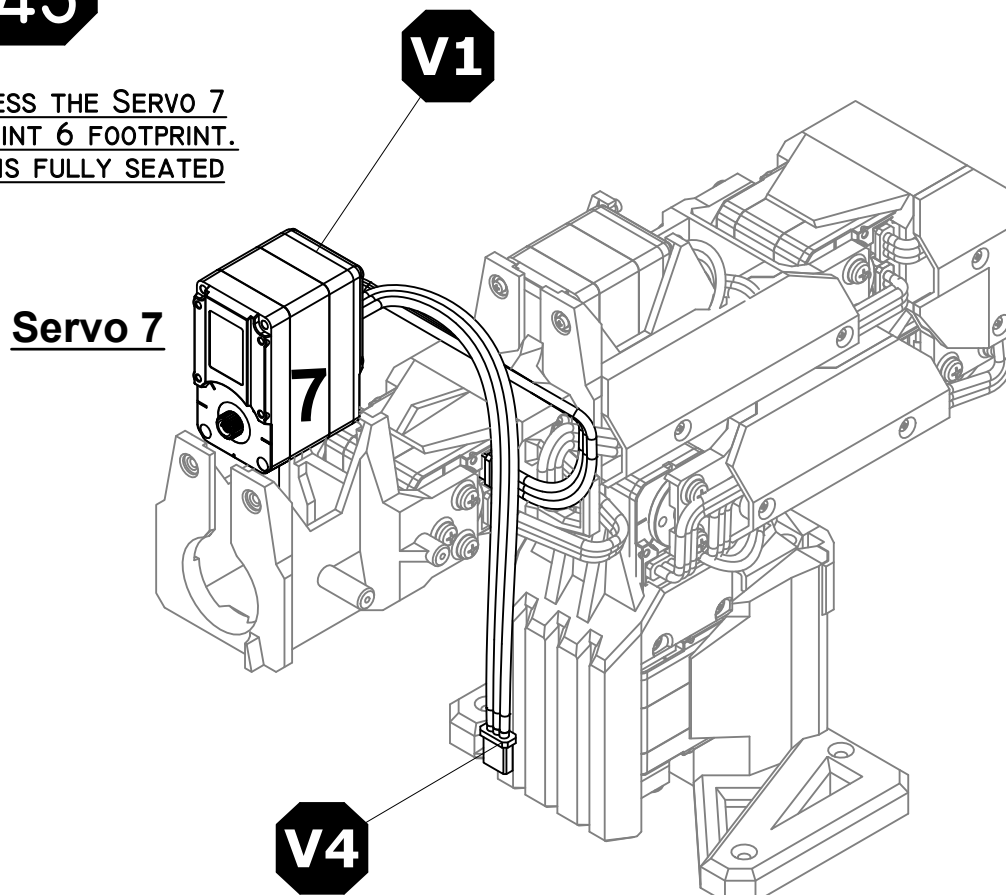
SERVO WIRES



V4 **1x**

STEP 45

GENTLY PRESS THE SERVO 7
INTO THE JOINT 6 FOOTPRINT.
ENSURE IT IS FULLY SEATED



ELROBOT

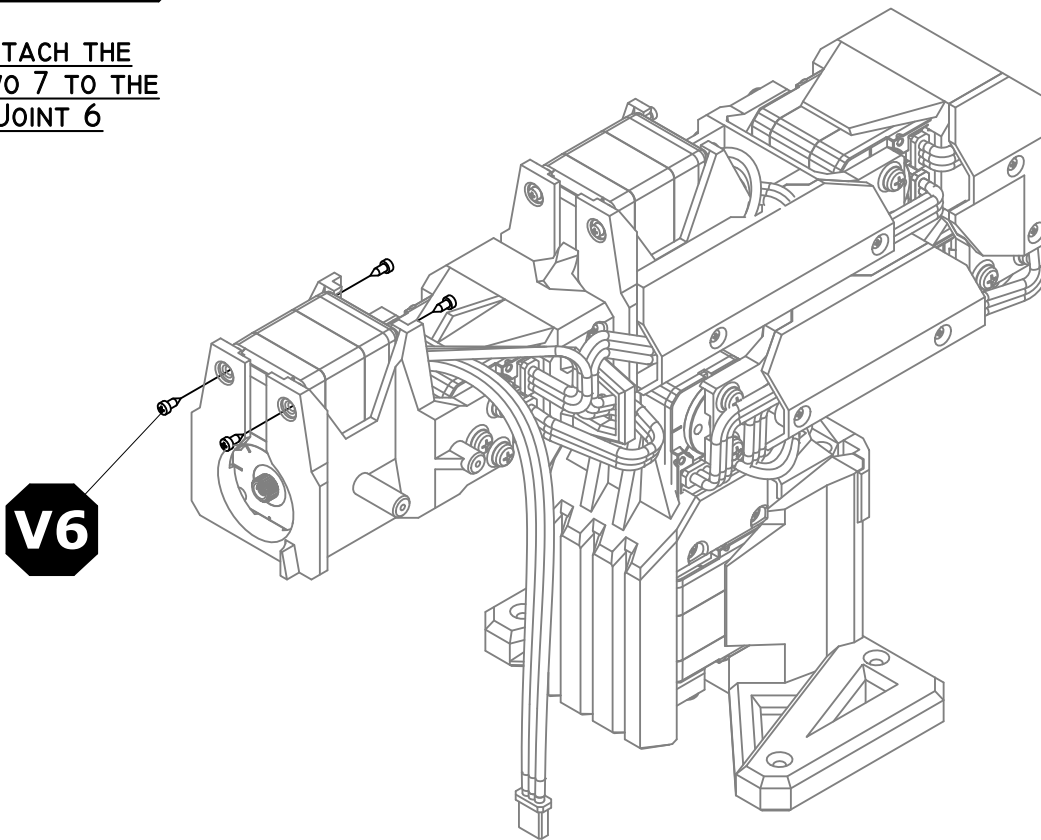
STEP 46

ATTACH THE
SERVO 7 TO THE
JOINT 6

SELF-TAPPING
SCREW M2X6

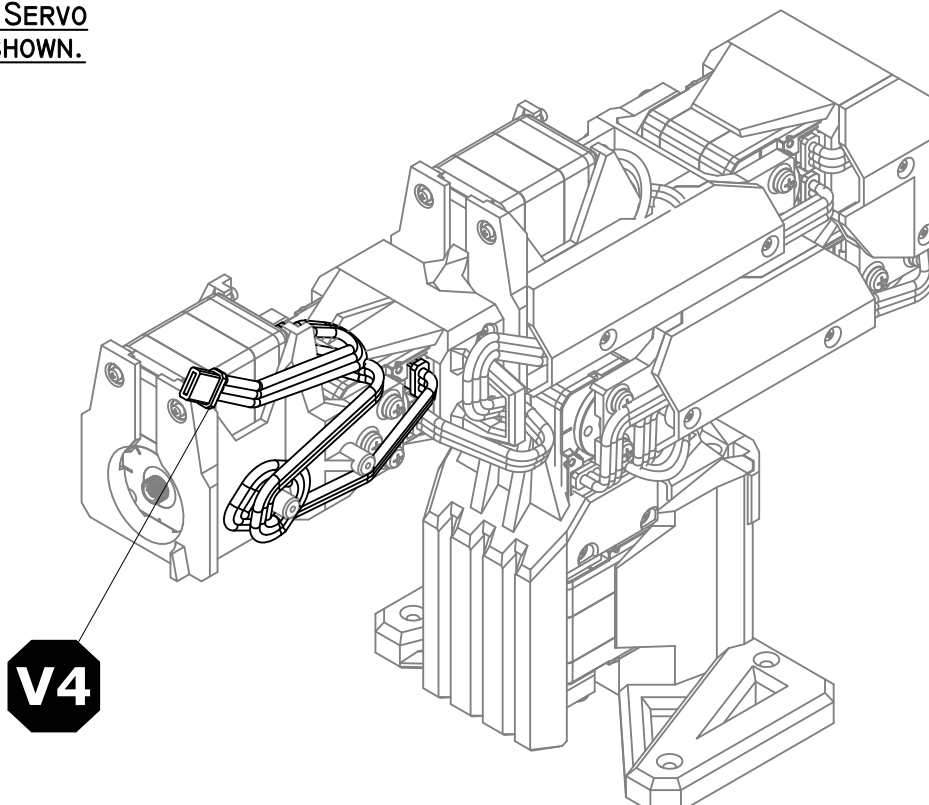


V6 4x



STEP 47

ROUTE THE SERVO
WIRES AS SHOWN.

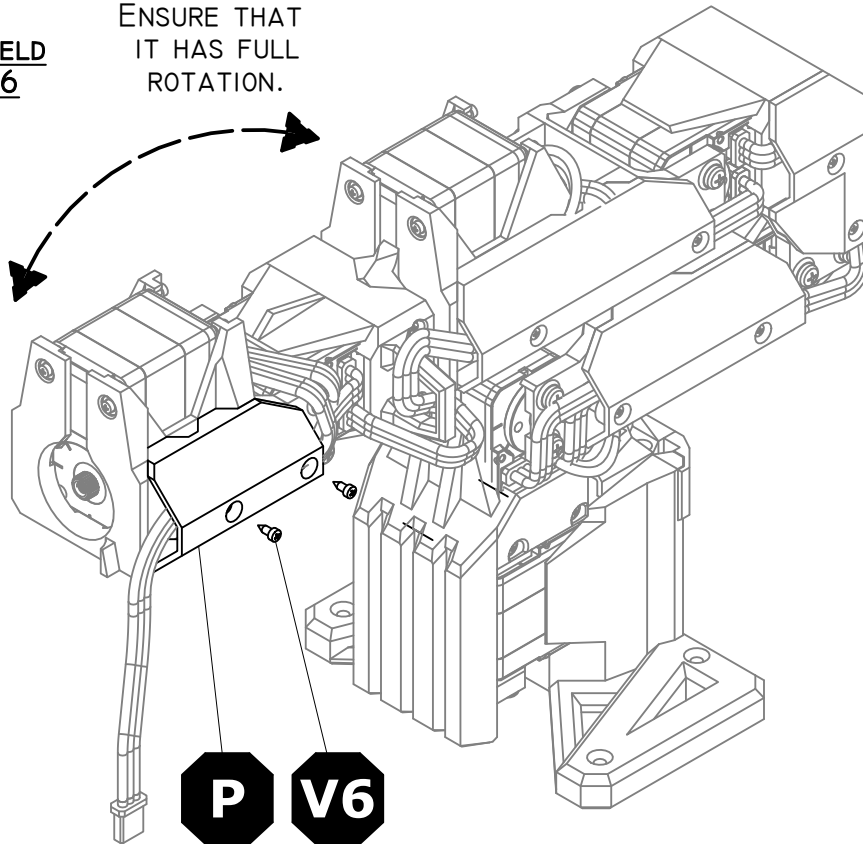


ELROBOT

STEP 48

ATTACH THE SHIELD
TO THE JOINT 6

ENSURE THAT
IT HAS FULL
ROTATION.



JOINT 6 SHIELD

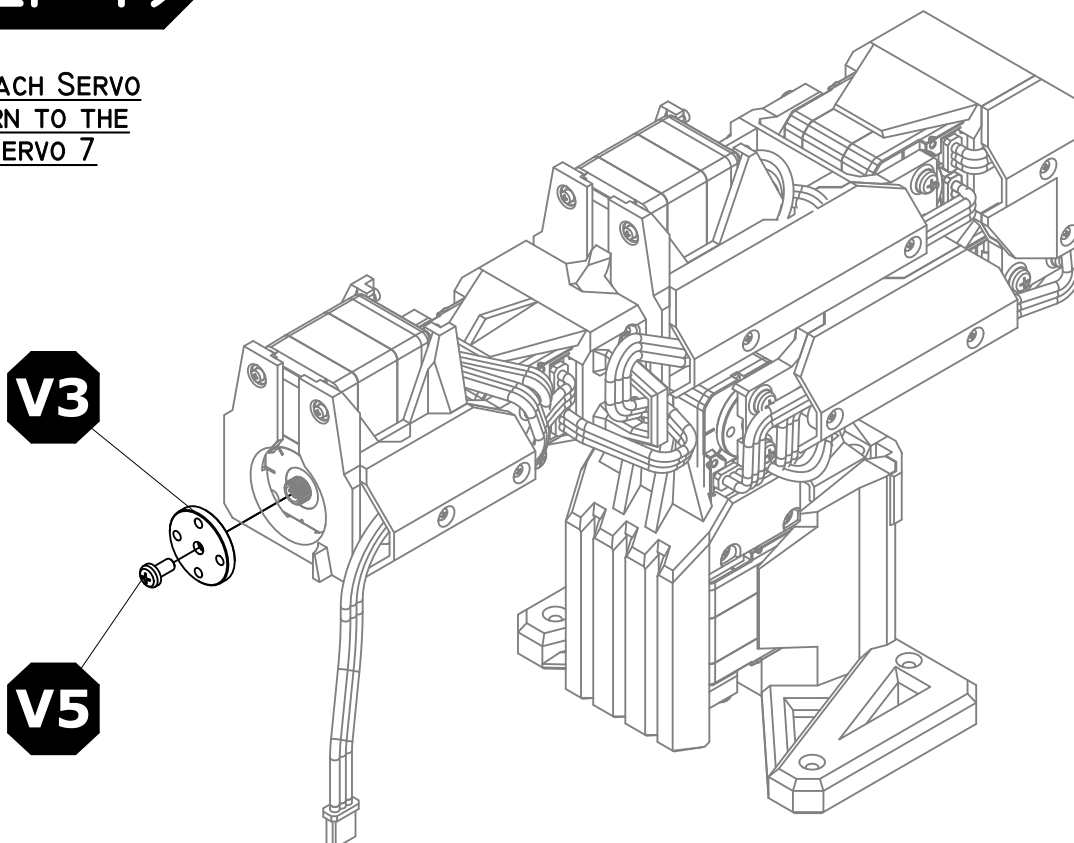
P **1x**

SELF-TAPPING
SCREW M2x6

V6 **2x**

STEP 49

ATTACH SERVO
HORN TO THE
SERVO 7



SERVO HORN

V3 **1x**

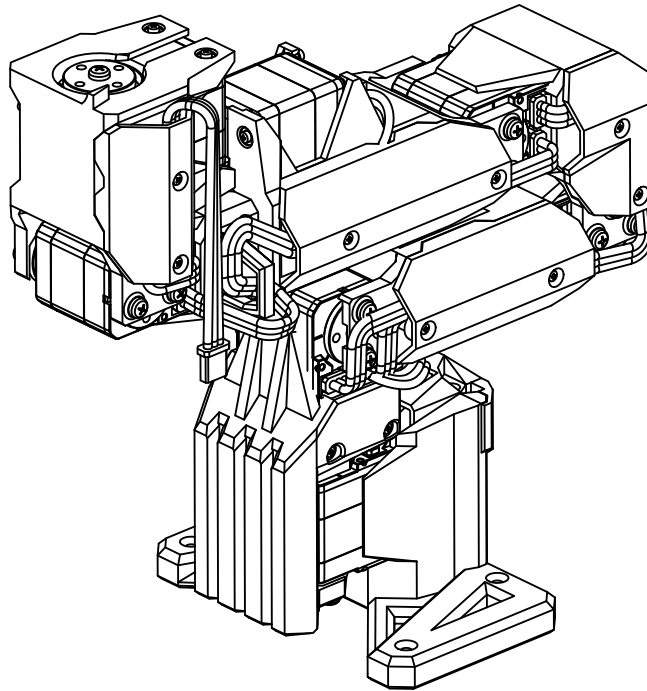
SCREW M3x8

V5 **1x**

ELROBOT

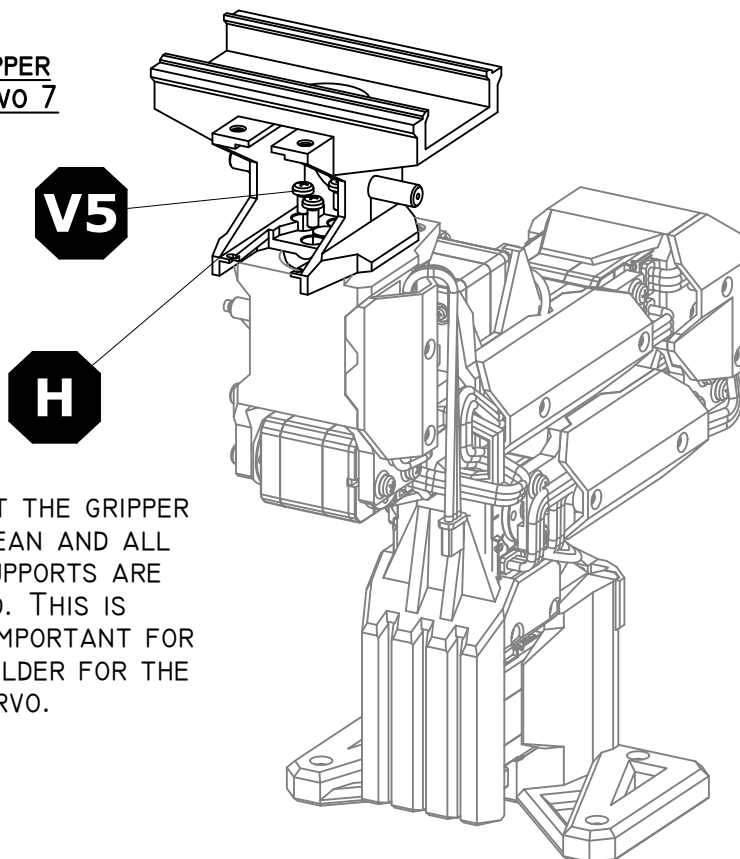
STEP 50

TURN ELROBOT
LIKE SHOWN



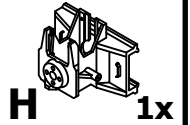
STEP 51

ATTACH THE GRIPPER
BASE TO THE SERVO 7



ENSURE THAT THE GRIPPER
BASE IS CLEAN AND ALL
PRINTED SUPPORTS ARE
REMOVED. THIS IS
ESPECIALLY IMPORTANT FOR
THE PLACEHOLDER FOR THE
SERVO.

GRIPPER BASE



H 1x

SCREW M3x8

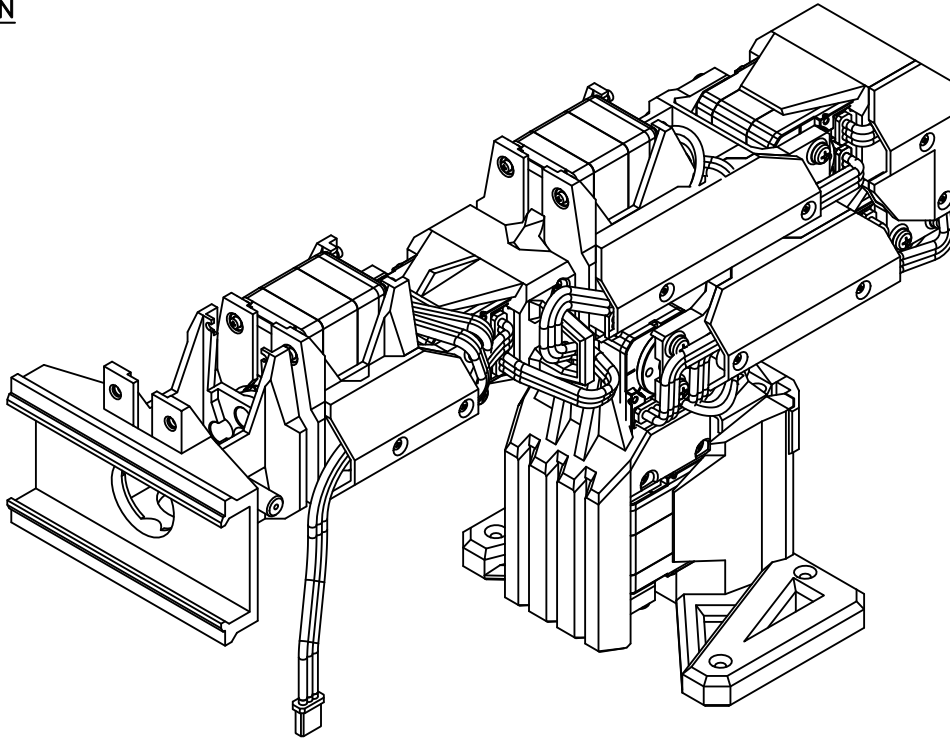


V5 4x

ELROBOT

STEP 52

TURN ELROBOT
LIKE SHOWN



STEP 53

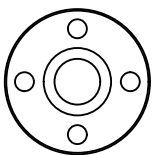
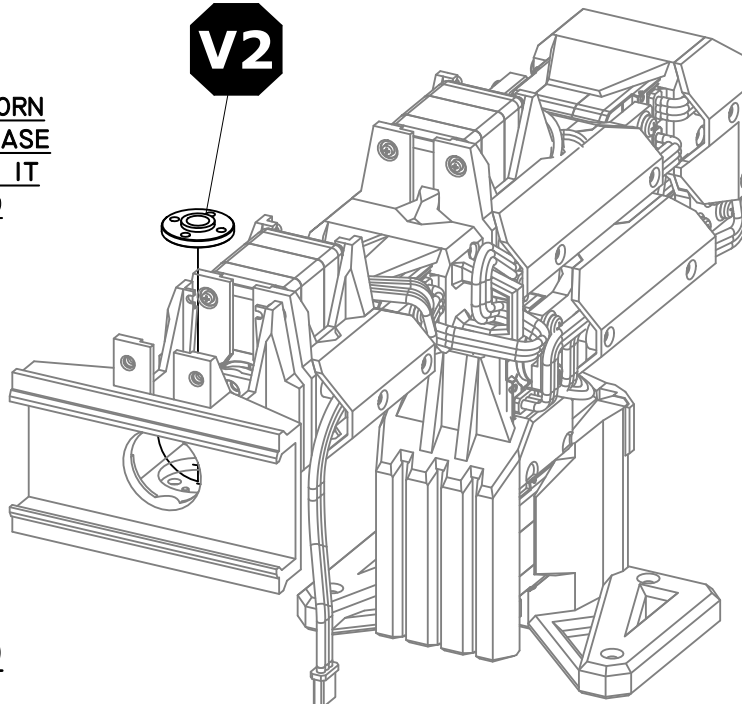
PRESS THE IDLER HORN
INTO THE GRIPPER BASE
FOOTPRINT. ENSURE IT
IS FULLY SEATED

V2

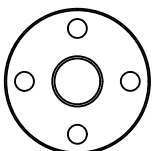
IDLER HORN



V2 1x



Top



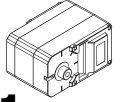
Bottom

ELROBOT

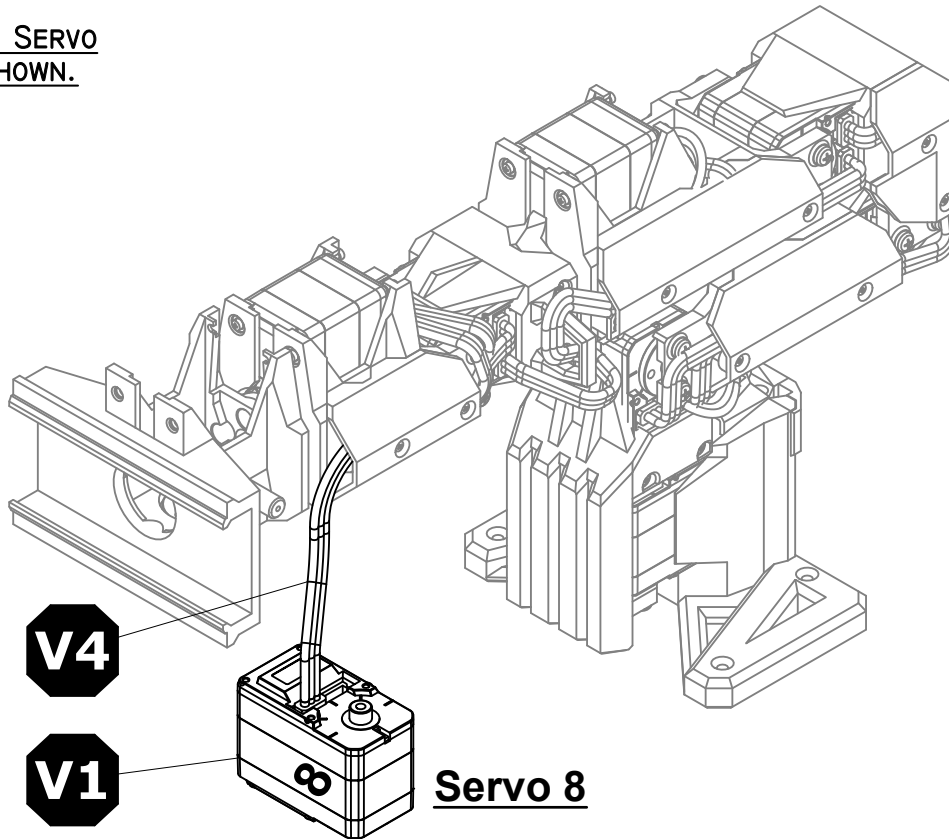
STEP 54

CONNECT THE SERVO
WIRES AS SHOWN.

SERVO ST3215

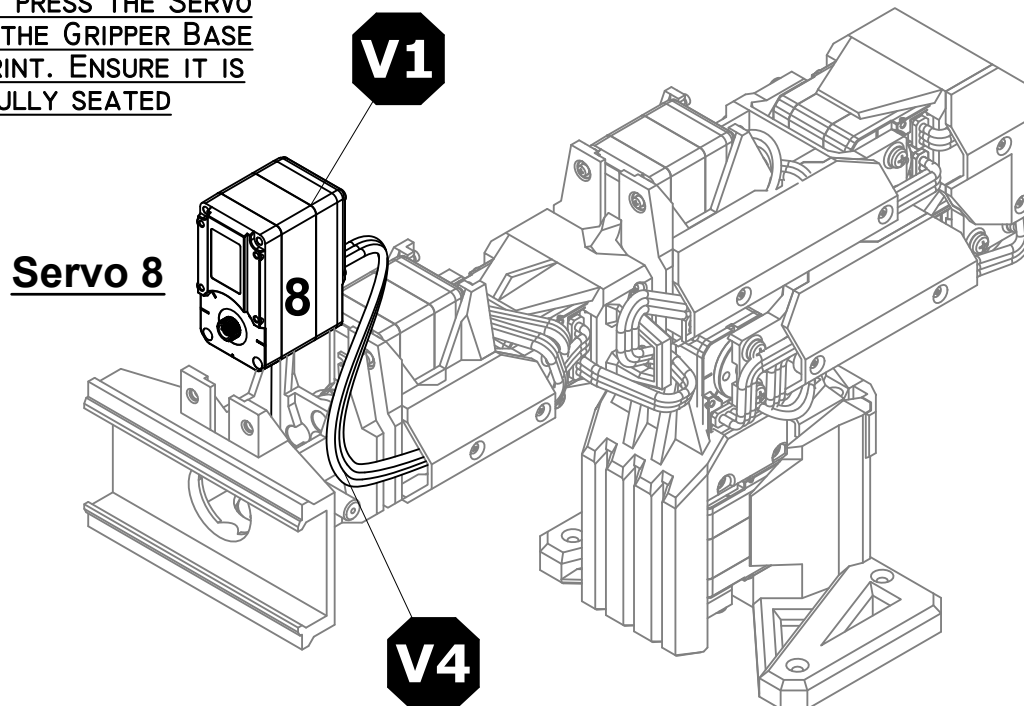


V1 1x



STEP 55

GENTLY PRESS THE SERVO
8 INTO THE GRIPPER BASE
FOOTPRINT. ENSURE IT IS
FULLY SEATED

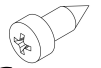


ELROBOT

STEP 56

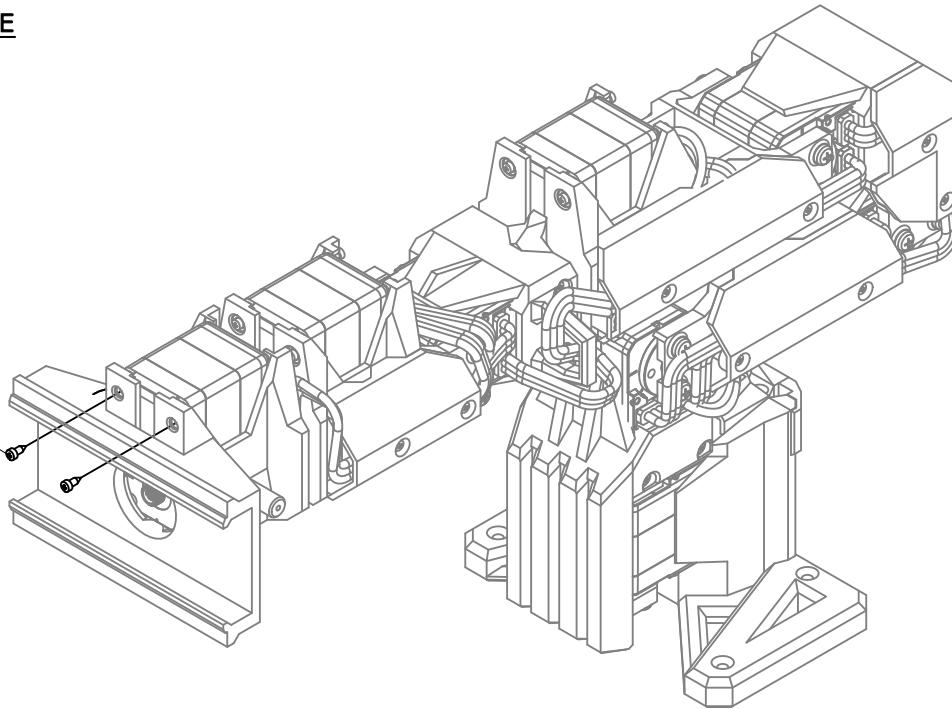
ATTACH THE
SERVO 8 TO THE
GRIPPER BASE

SELF-TAPPING
SCREW M2X6



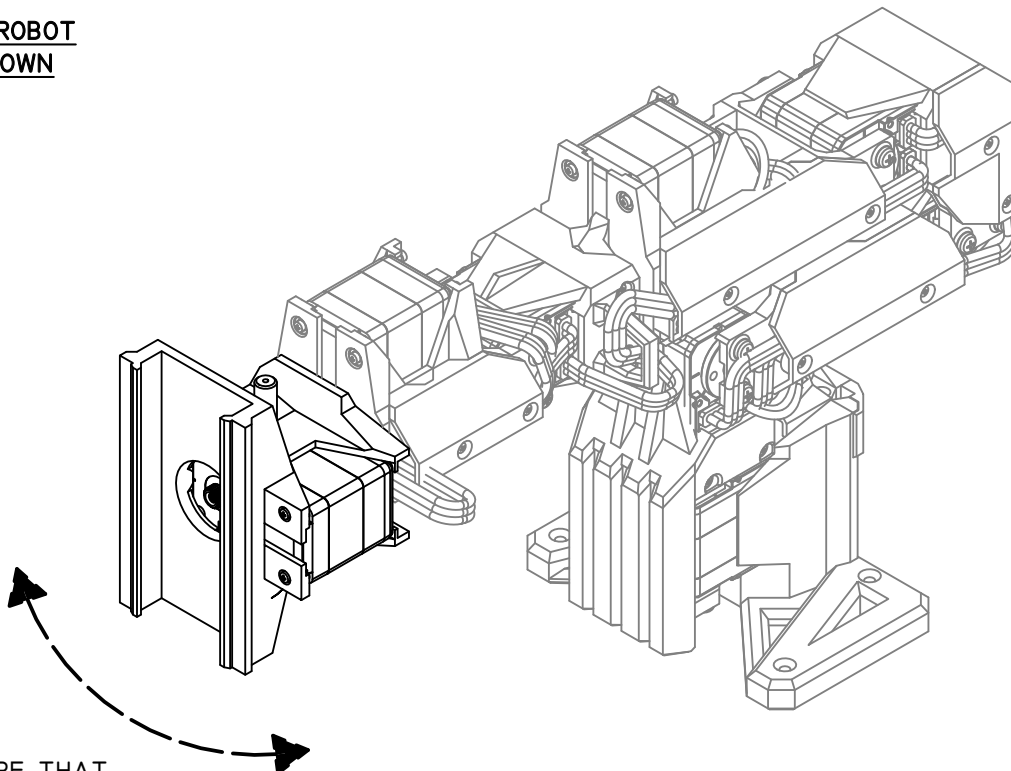
V6 2x

V6



STEP 57

TURN ELROBOT
LIKE SHOWN

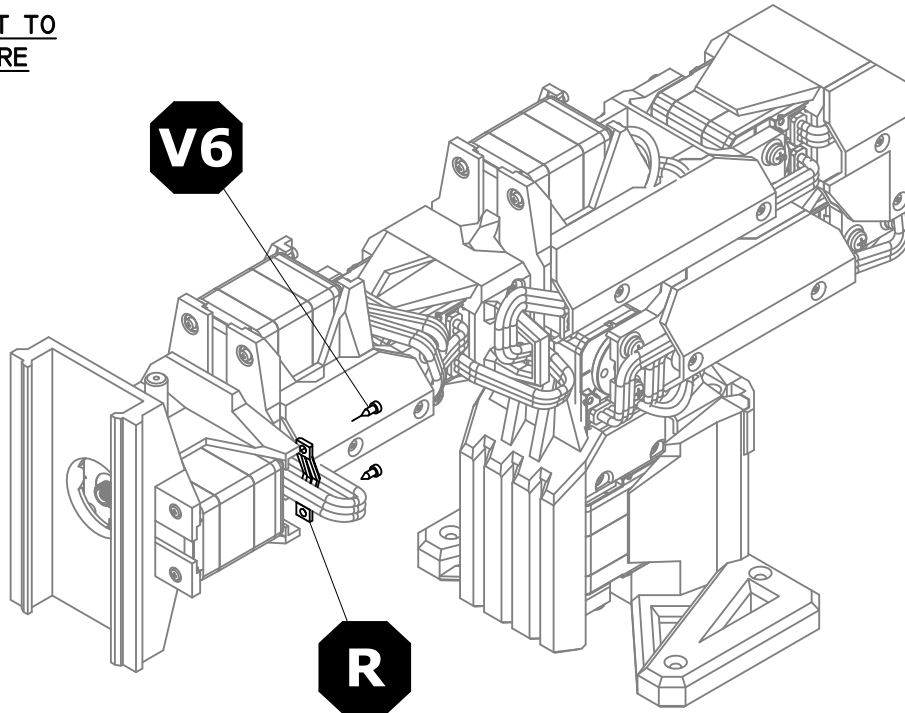


ENSURE THAT
IT HAS FULL
ROTATION.

ELROBOT

STEP 58

USE THE BRACKET TO
SECURE THE WIRE



WIRING BRACKET



R **1x**

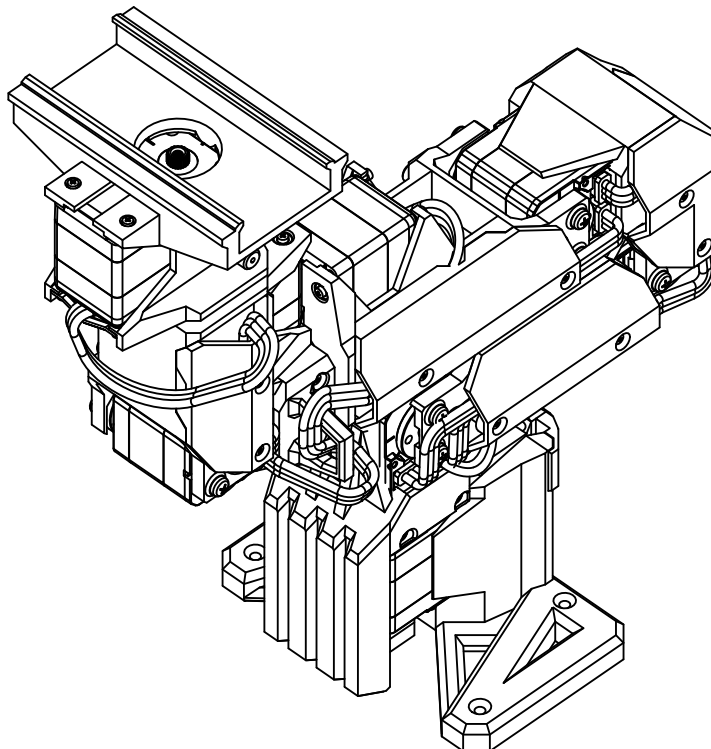
SELF-TAPPING
SCREW M2x6



V6 **2x**

STEP 59

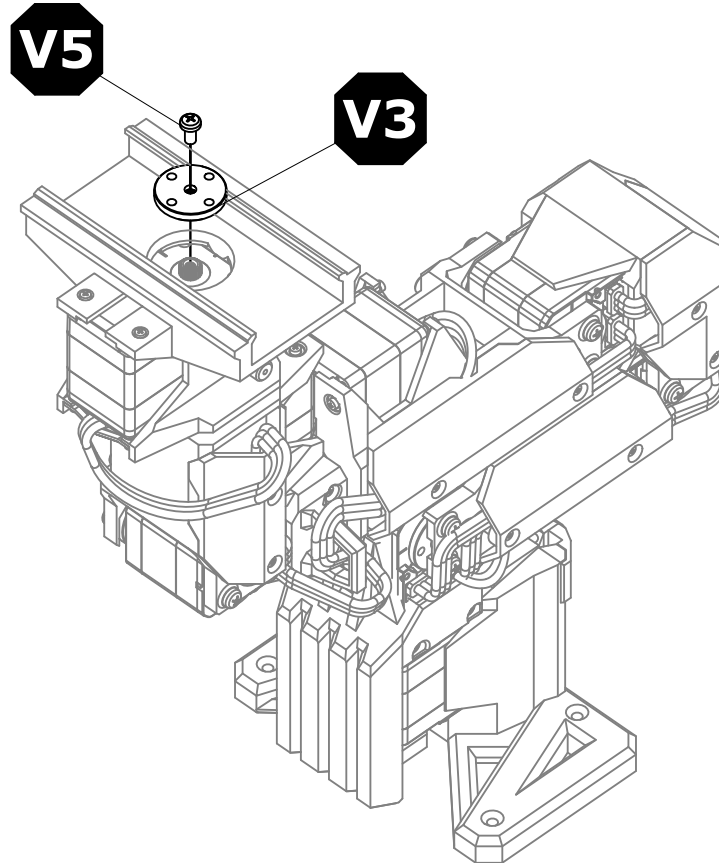
TURN ELROBOT
LIKE SHOWN



ELROBOT

STEP 60

ATTACH SERVO HORN
TO THE SERVO 8



SERVO HORN



V3 1x

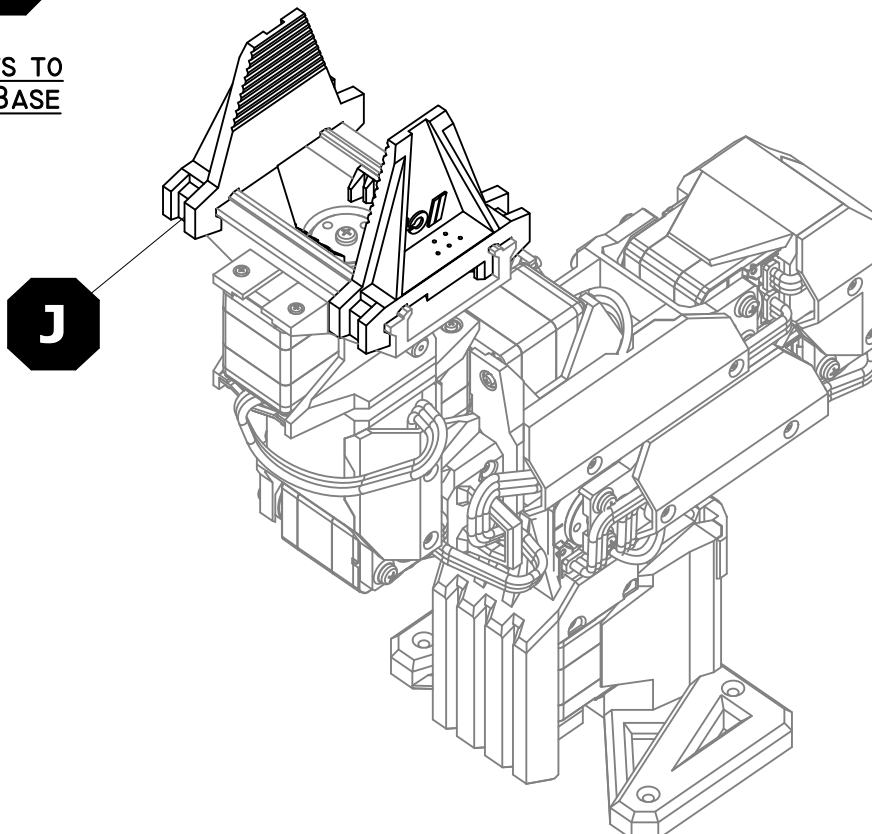
SCREW M3x8



V5 1x

STEP 61

RAIL THE JAWS TO
THE GRIPPER BASE



GRIPPER JAW



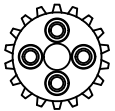
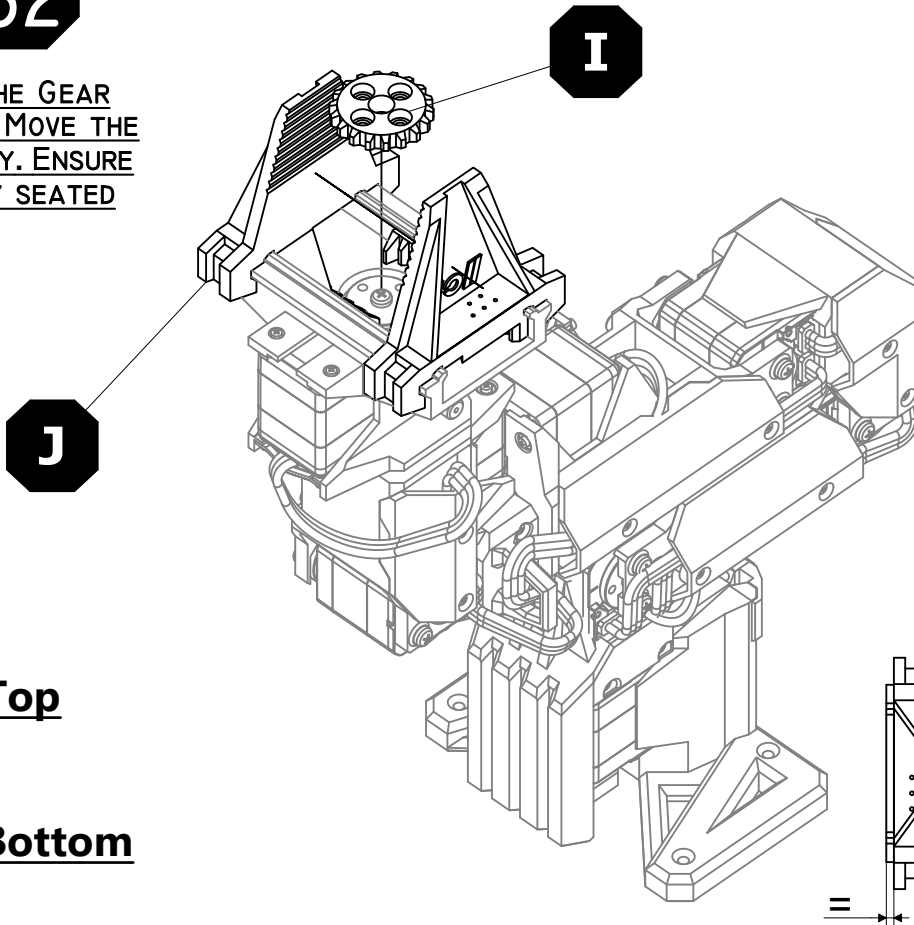
J 2x

ELROBOT

STEP 62

INSTALL THE GEAR
LIKE SHOWN. MOVE THE
JAWS EVENLY. ENSURE
IT IS FULLY SEATED

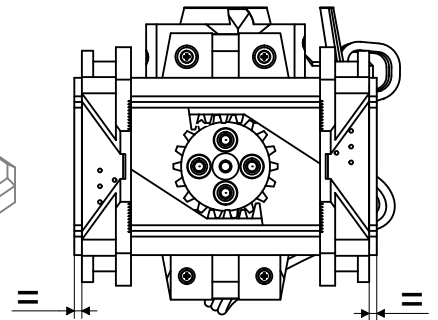
GRIPPER GEAR



Top



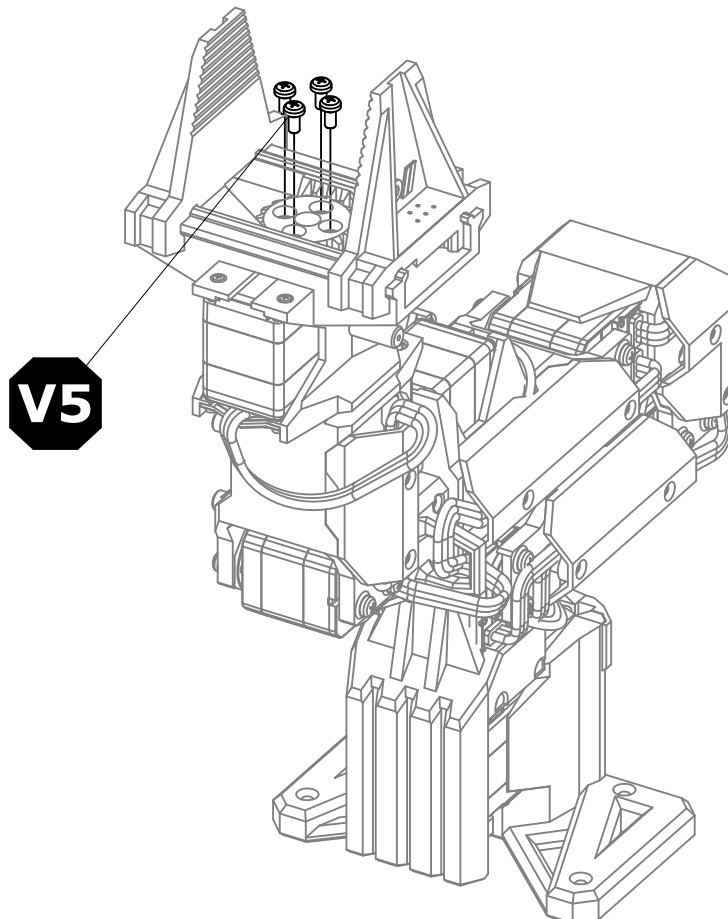
Bottom



STEP 63

ATTACH THE GEAR
TO THE SERVO 8

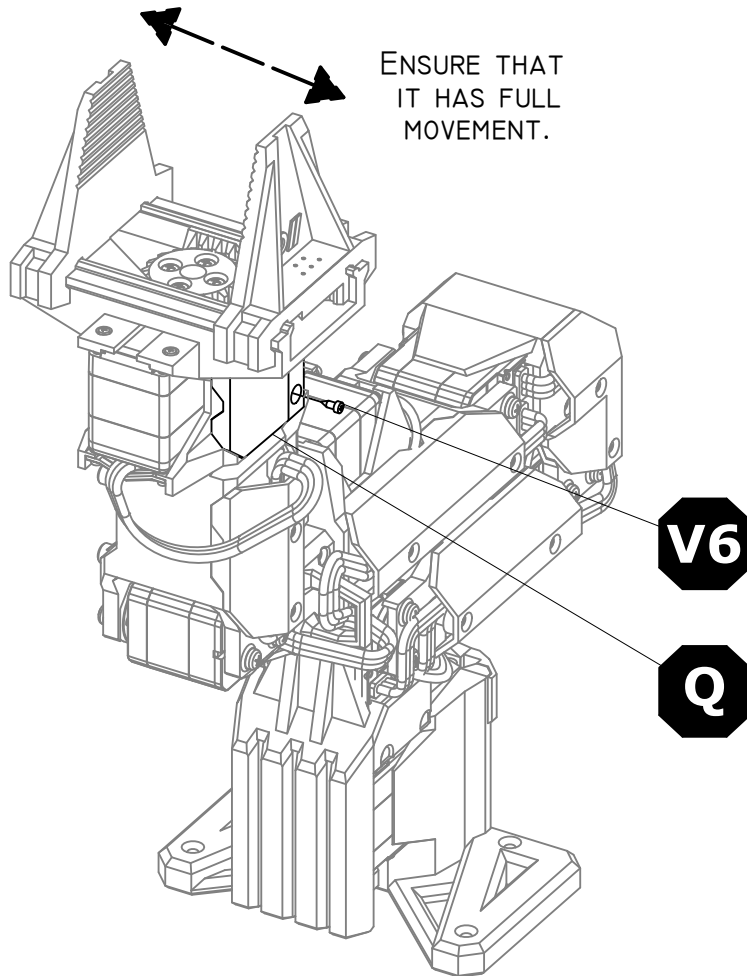
SCREW M3x8



ELROBOT

STEP 64

ATTACH THE SHIELD TO
THE GRIPPER BASE



GRIPPER BASE
SHIELD

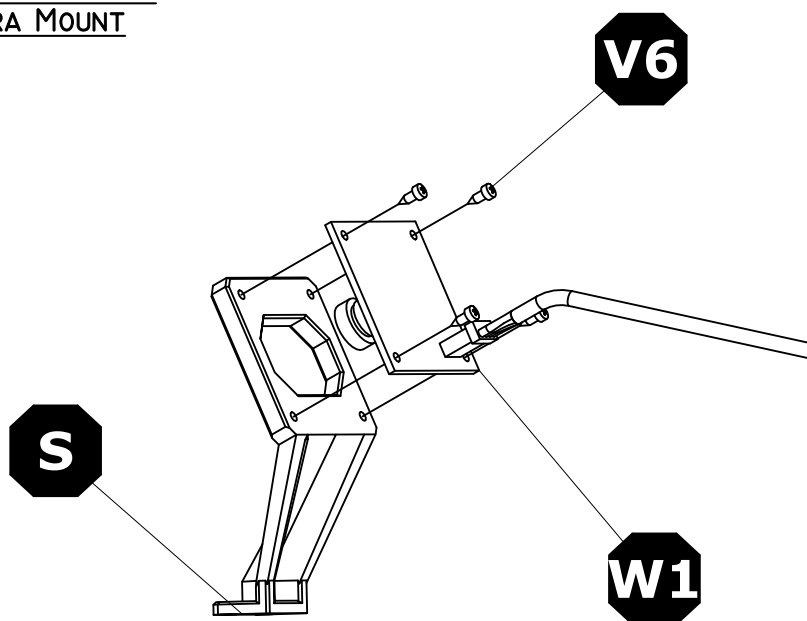
Q **1x**

SELF-TAPPING
SCREW M2x6

V6 **1x**

STEP 65

ATTACH THE CAMERA TO
THE CAMERA MOUNT



CAMERA MOUNT

S **1x**

SELF-TAPPING
SCREW M2x6

V6 **4x**

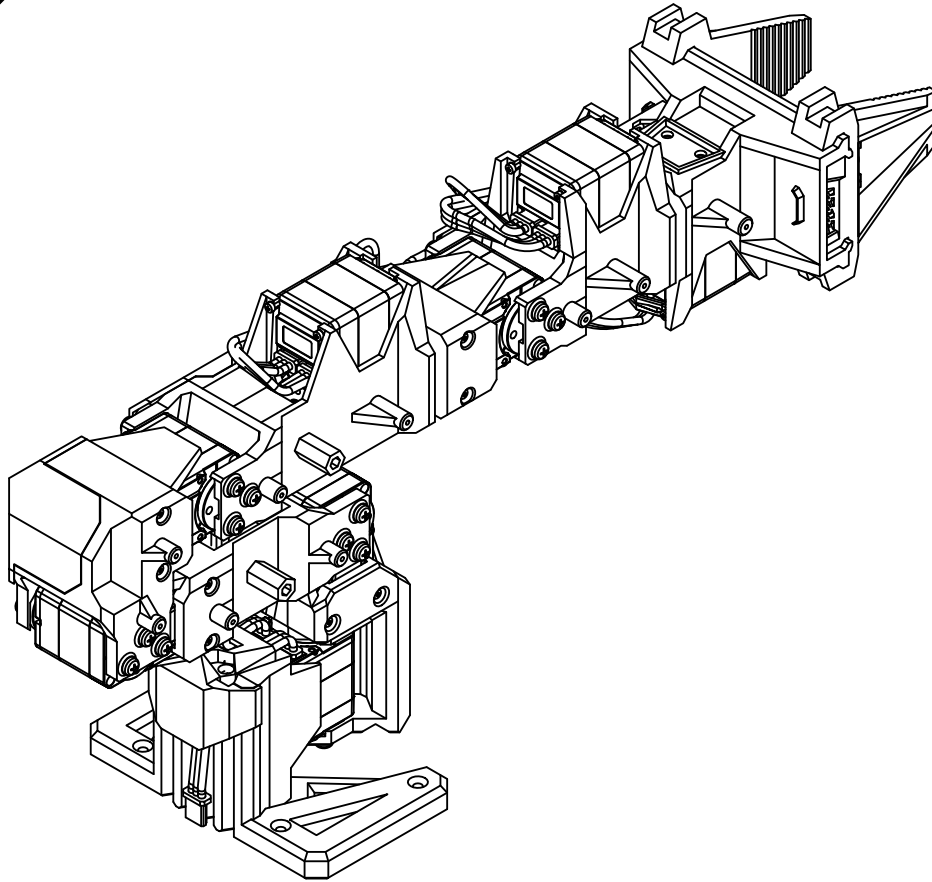
UVC CAMERA
INNOMAKER U20CAM

W1 **1x**

ELROBOT

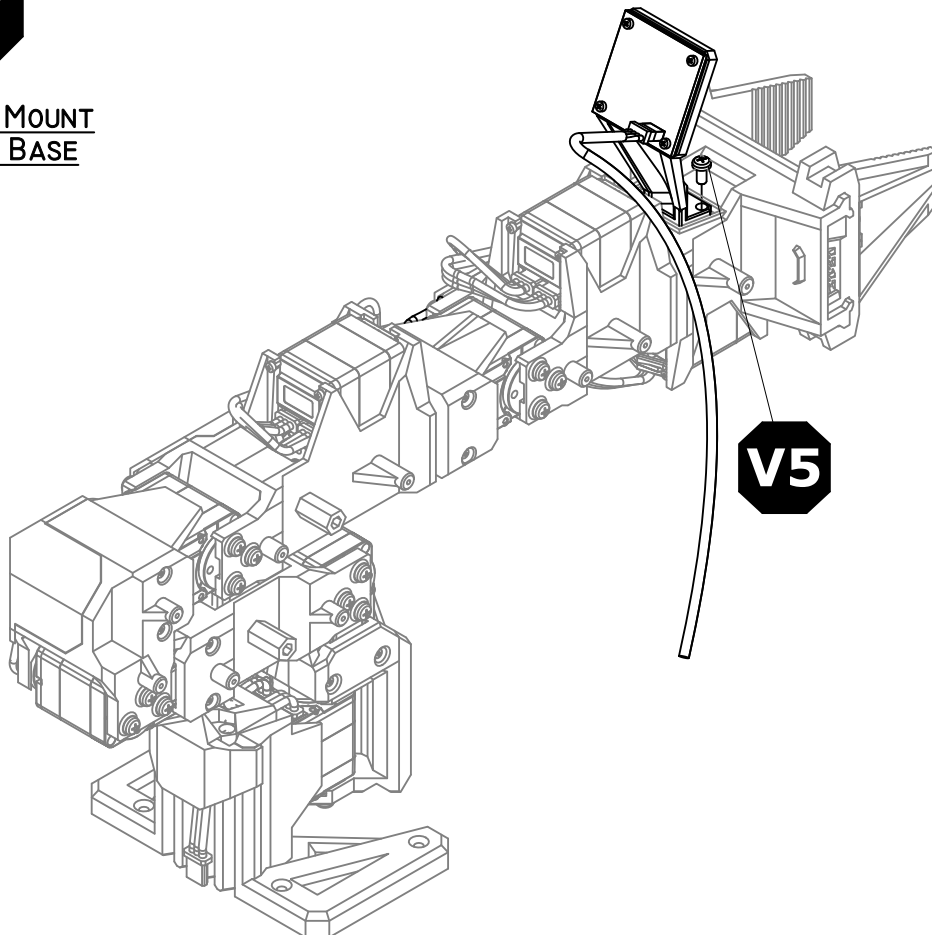
STEP 66

TURN ELROBOT
LIKE SHOWN



STEP 67

ATTACH CAMERA MOUNT
TO THE GRIPPER BASE



SCREW M3x8

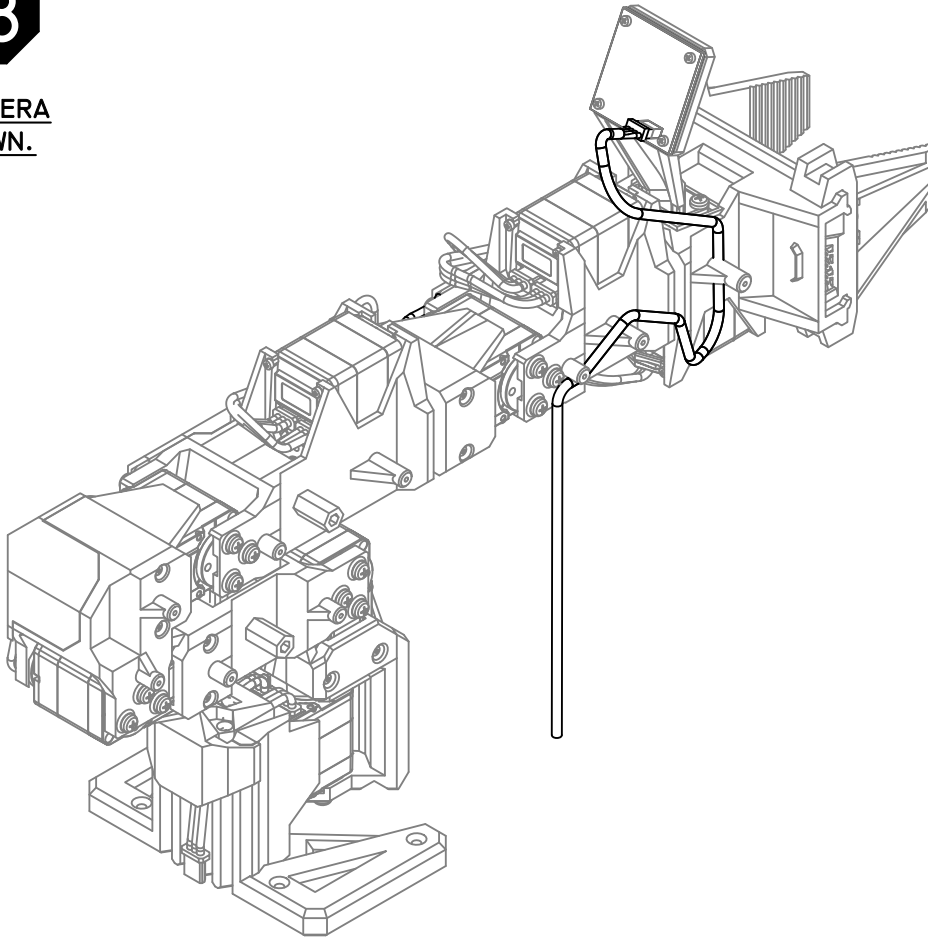


V5 2x

ELROBOT

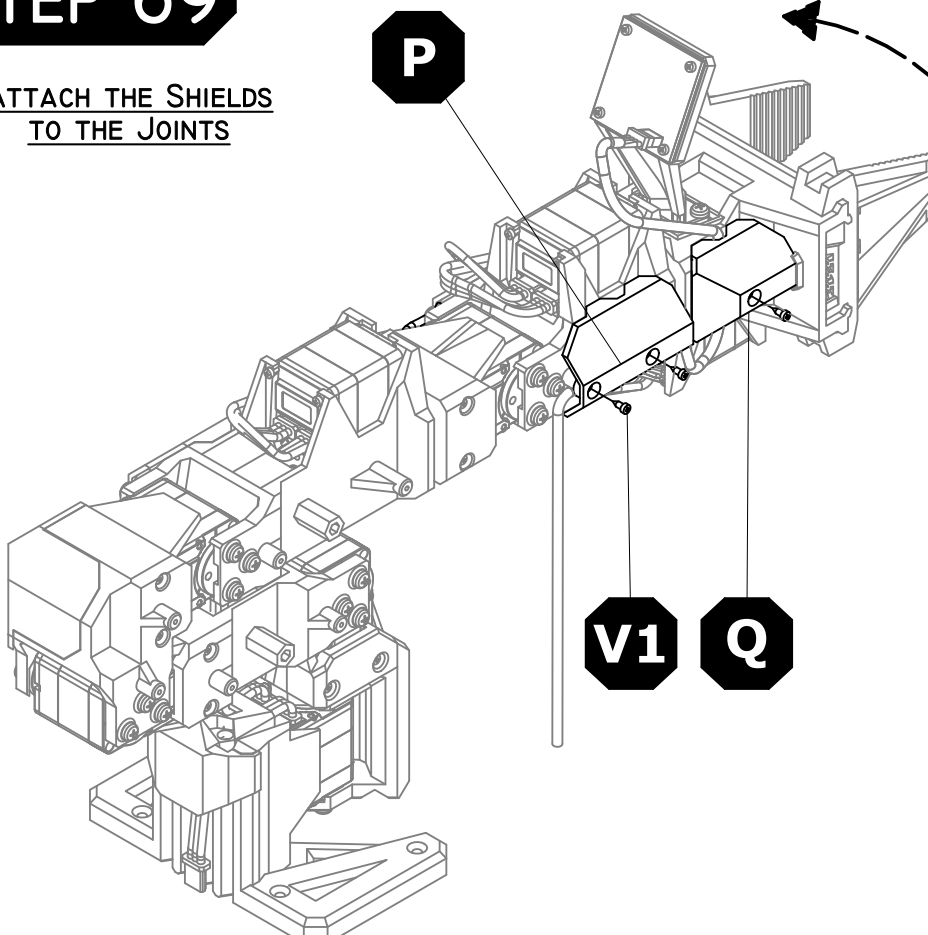
STEP 68

ROUTE THE CAMERA
WIRES AS SHOWN.



STEP 69

ATTACH THE SHIELDS
TO THE JOINTS



ENSURE THAT
IT HAS FULL
ROTATION.

JOINT 6 SHIELD



P 1x

GRIPPER BASE
SHIELD



Q 1x

SELF-TAPPING
SCREW M2X6

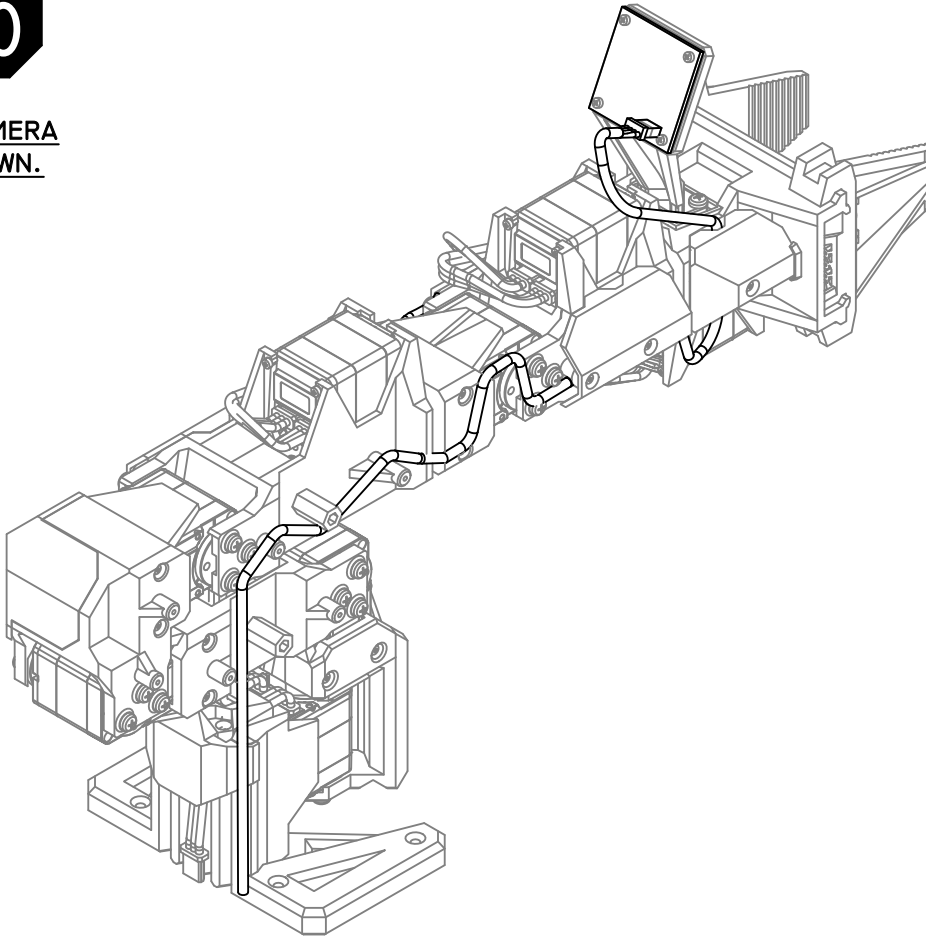


V6 3x

ELROBOT

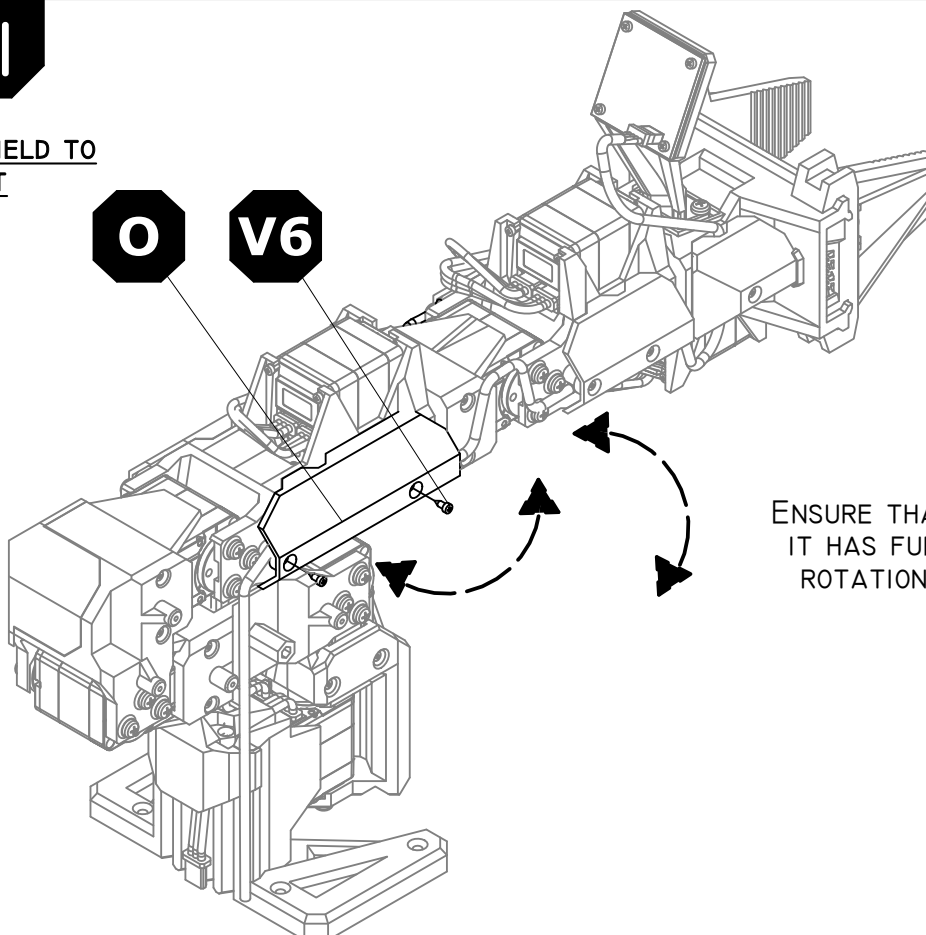
STEP 70

ROUTE THE CAMERA
WIRES AS SHOWN.



STEP 71

ATTACH THE SHIELD TO
THE JOINT



JOINT 4 SHIELD

O **1x**

SELF-TAPPING
SCREW M2x6

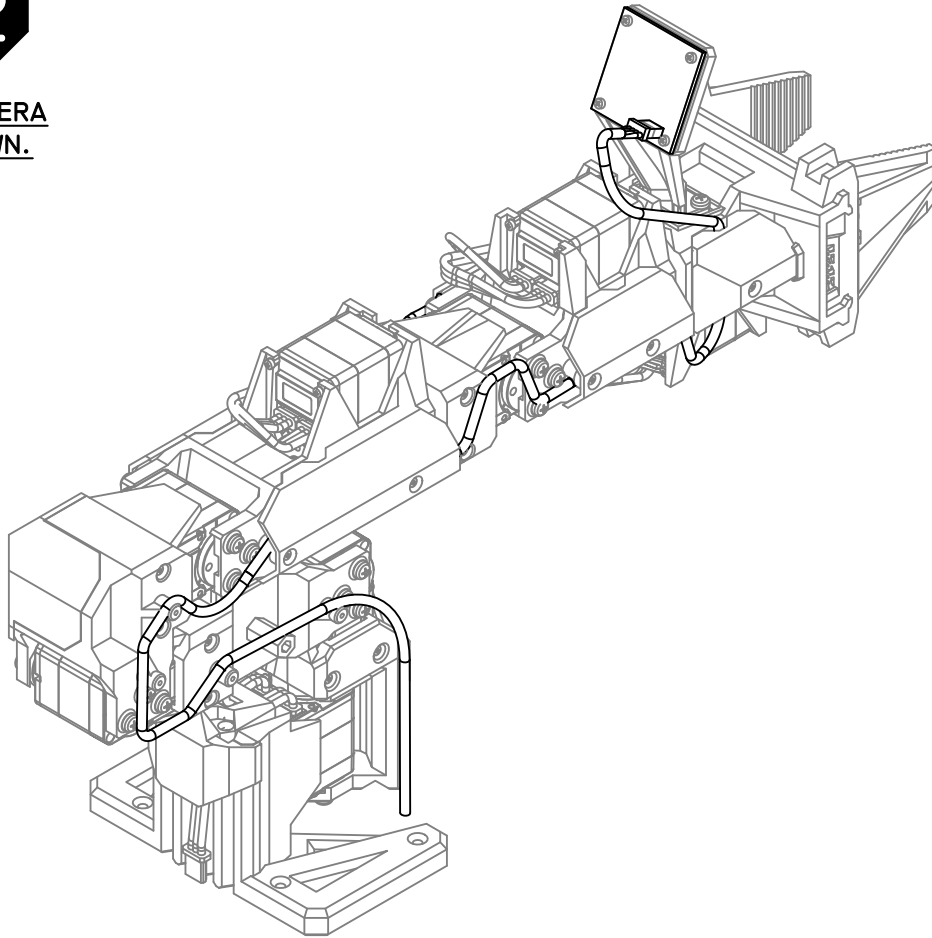
V6 **2x**

ENSURE THAT
IT HAS FULL
ROTATION.

ELROBOT

STEP 72

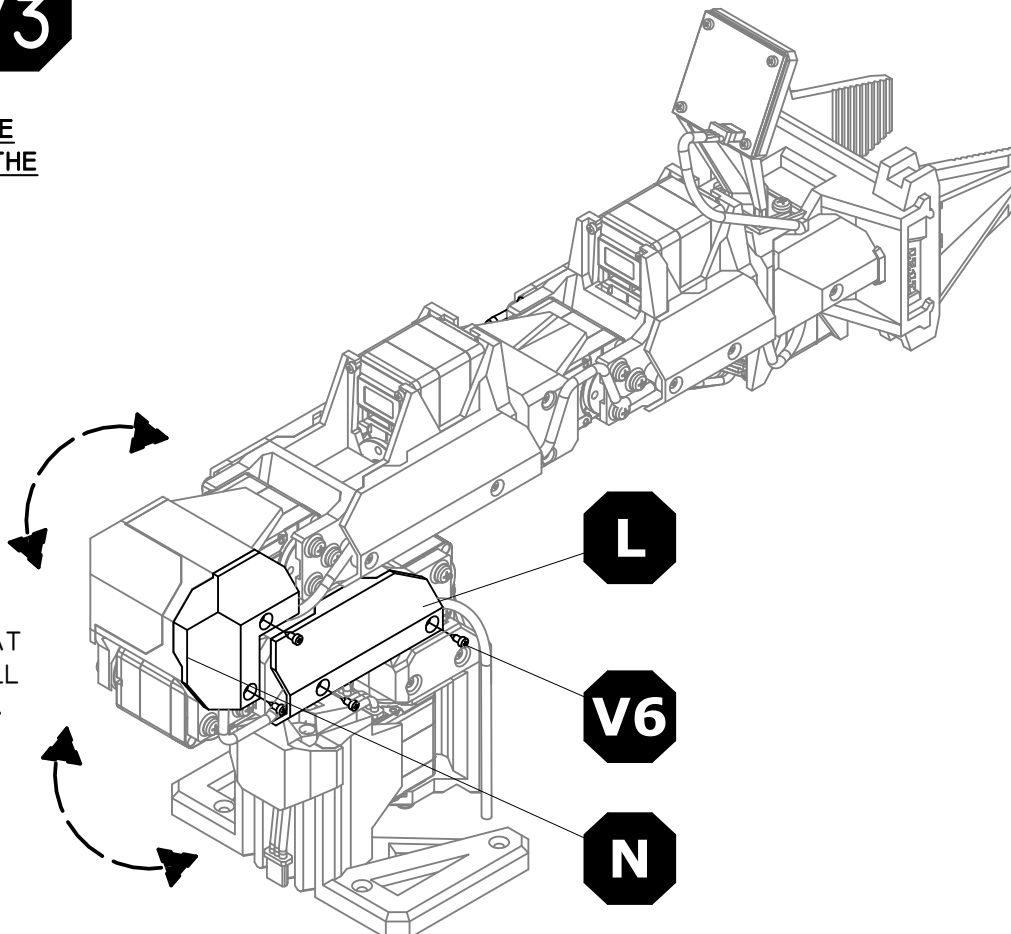
ROUTE THE CAMERA
WIRES AS SHOWN.



STEP 73

ATTACH THE
SHIELDS TO THE
JOINTS

ENSURE THAT
IT HAS FULL
ROTATION.



JOINT 2 SHIELD



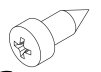
L **1x**

JOINT 2 SHIELD
DRIVE SIDE



N **1x**

SELF-TAPPING
SCREW M2X6

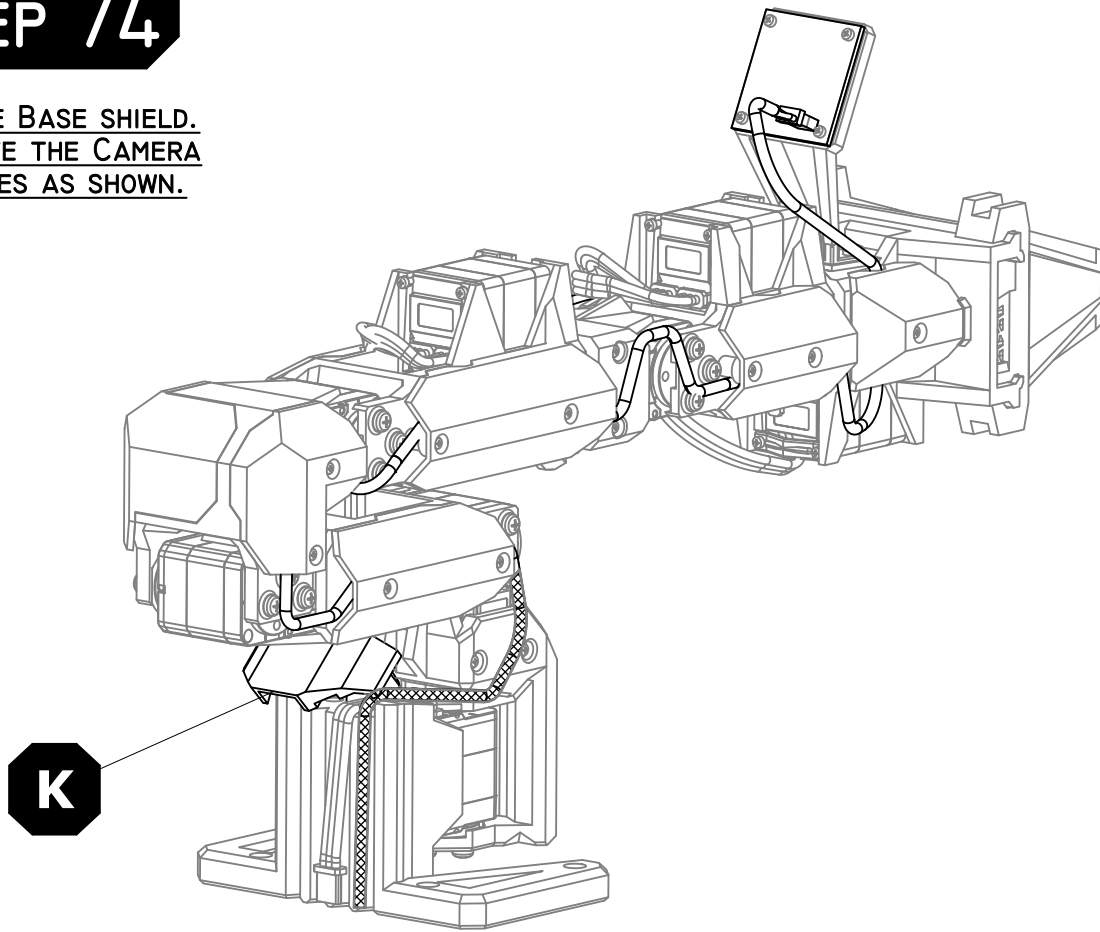


V6 **4x**

ELROBOT

STEP 74

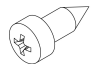
MOVE BASE SHIELD.
ROUTE THE CAMERA
WIRES AS SHOWN.



STEP 75

ATTACH THE BASE
SHIELD TO THE BASE

SELF-TAPPING
SCREW M2x6



V6 2x

